Supply Chain Analytics 12 (2025) 100154

Contents lists available at ScienceDirect

Supply Chain Analytics

journal homepage: www.sciencedirect.com/journal/supply-chain-analytics

Full length article

A comparative study of multi-algorithm optimization for inventory analytics in supply chains

Oussama Zabraoui *, Yahya Hmamou[1], Anas Chafi[1], Salaheddine Kammouri Alami[1]

The Industrial Techniques Laboratory, Faculty of Science and Technology, Sidi Mohammed Ben Abdellah University, B.P. 2202 Route d'Imouzzer, FEZ, 30050, Morocco

ARTICLE INFO

Keywords:
Supply chain optimization
Demand forecasting
Inventory analytics
Deep learning
Hybrid optimization
Reinforcement learning

ABSTRACT

Effective management of inventory is essential for achieving high service levels, minimizing costs, and maintaining the overall resilience of retail supply chains—particularly in complex, real-world environments. Conventional strategies often prove inadequate because they rely on rigid assumptions or single-technique models that fail to accommodate practical challenges such as fluctuating demand, unpredictable lead times, and disruptions in supply.

To bridge this gap, our research undertakes a comprehensive comparison of multiple approaches — including Reinforcement Learning (RL), Genetic Algorithms (GA), Deep Learning (DL), Machine Learning (ML), and heuristic techniques — evaluated within a consistent and realistic testing framework based on the Walmart M5 dataset. This dataset offers a robust benchmark, containing multi-store, multi-item sales data that captures seasonal trends, event-driven demand variations, and price sensitivity. We introduce and evaluate an innovative hybrid methodology that combines a Genetic Algorithm with a Deep Q-Network (GA–DQN). The GA component conducts a broad, global search to optimize static inventory parameters such as reorder points and safety stock, while the DQN module learns adaptive, state-aware ordering strategies that can respond to dynamic, uncertain conditions. Our results show that this hybrid GA–DQN model achieves a significant improvement over a standalone DQN baseline—raising the service level from 61% to 94% and simultaneously lowering overall inventory costs. The framework we propose is modular and includes three key components: demand forecasting using Long Short-Term Memory (LSTM) networks to capture temporal sales patterns; GA-based optimization to fine-tune static policy parameters; and RL-driven adaptive control to support responsive, real-time ordering decisions. This integrated approach delivers a scalable, data-driven solution well-suited to the demands of modern retail supply chains, effectively addressing issues such as supplier unreliability, demand uncertainty, and the management of perishable goods.

1. Introduction

Inventory management is a cornerstone of supply chain operations, directly impacting operational efficiency, customer satisfaction, and financial performance [1]. Effective inventory management ensures optimal stock levels to meet variable customer demand while minimizing costs associated with overstocking (holding costs), understocking (stockout costs), and frequent ordering (ordering costs) [2]. However, modern supply chains face unprecedented complexity due to real-time demand volatility, uncertain lead times, supplier variability, and multi-echelon interactions [3]. Traditional inventory models, such as Economic Order Quantity (EOQ), (s, S) policies, and ABC classification, rely on static assumptions like constant demand and fixed lead times, which are increasingly inadequate in dynamic, data-driven environments [1,4]. These limitations lead to suboptimal inventory decisions, elevated costs, and diminished service levels, particularly in sectors like retail, manufacturing, and e-commerce [5].

The emergence of artificial intelligence (AI) and machine learning (ML) has transformed inventory management by enabling adaptive, scalable, and data-driven solutions [6]. Reinforcement Learning (RL) has gained prominence for its ability to generate dynamic inventory policies by modeling inventory control as a Markov Decision Process (MDP). RL algorithms, such as Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO), learn optimal actions (e.g., order quantities) by maximizing a reward function that balances holding, stockout,

* Corresponding author.
E-mail addresses: oussama.zabraoui@umba.ac.ma (O. Zabraoui), yahyahmamou@usmba.ac.ma (Y. Hmamou), anaschafi@gmail.com (A. Chafi), salaheddine.kammourialami@usmba.ac.ma (S.K. Alami).
[1] Contributing authors.

https://doi.org/10.1016/j.sca.2025.100154
Received 6 June 2025; Received in revised form 2 August 2025; Accepted 8 August 2025
Available online 21 August 2025
2949-8635/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

and ordering costs [7]. RL's strength lies in its adaptability to demand uncertainty and multi-echelon supply chain dynamics, making it ideal for complex scenarios [8]. However, RL's computational intensity and reliance on well-defined reward functions necessitate careful implementation [3].

Genetic Algorithms (GA), a class of metaheuristics, provide an alternative approach by optimizing static inventory parameters such as reorder points (ROP), safety stock levels, and fixed order quantities. GA evolves a population of candidate solutions through selection, crossover, and mutation to minimize total inventory costs while meeting service level constraints [5]. GA's versatility in handling multi-product and multi-objective optimization problems makes it particularly effective in supply chains with diverse product portfolios [9]. Recent studies have also explored hybrid GA frameworks to enhance convergence and solution quality [10].

Deep Learning (DL), particularly Long Short-Term Memory (LSTM) networks, has revolutionized demand forecasting by capturing temporal dependencies and non-linear patterns in time-series data [11]. Unlike traditional ML models like Random Forests or Gradient Boosting, which are effective for short-term predictions, LSTM models excel in long-term forecasting by leveraging sequential data, enabling proactive inventory planning [12]. Integrating DL with optimization techniques like RL and GA enhances decision-making by providing accurate demand forecasts as inputs for policy generation and parameter tuning [13].

Notwithstanding, existing scholarship has considered the approaches of individual adaptation separately, but the direct comparison methods have not been conducted in their entirety as the present study takes into consideration approaches to each methodology under the same framework of benchmarking using realistic data. In earlier studies, synthetic data are often used, or the research focuses on the evaluation of one approach at a time, making it difficult to compare approaches to each other and limiting real-world implementation.

Despite these advancements, the literature lacks comprehensive comparative studies that evaluate RL, GA, DL, traditional ML, and heuristic methods within a unified framework. Most research focuses on individual techniques or synthetic datasets, limiting insights into their relative performance across real-world supply chain scenarios [4,6]. Furthermore, the application of these methods to large-scale retail datasets, such as the Walmart M5 dataset, remains underexplored, particularly in optimizing key performance indicators (KPIs) like total inventory cost, service level, stockout rate, order frequency, and bullwhip effect [7,8].

Section 3 explains the choice of the particular techniques. A brief sketch of every constitutive component — its aim, specification, and related measures of evaluation — are shown in Table 2. This design can be used to make a consistent, modular comparison of the whole pipeline.

This study addresses these gaps by conducting a systematic comparative analysis of RL, GA, DL, ML, and heuristic methods for inventory management using the Walmart M5 dataset, which provides granular sales, price, and event data across multiple stores and product categories. Our methodology integrates demand forecasting (via ML and DL), parameter optimization (via GA), and dynamic policy generation (via RL), benchmarked against heuristic baselines. The proposed architecture processes inputs such as product ID, date, quantity, price, lead time, and demand to output optimized order quantities, reorder points, and safety stock levels. By evaluating KPIs and conducting scenario analyses (e.g., demand spikes, supplier delays), we aim to identify the strengths and limitations of each method and provide actionable insights for practitioners.

This study offers three main contributions. First, we develop and assess an innovative inventory control strategy that integrates a Genetic Algorithm with a Deep Q-Network (GA–DQN). In this approach, the GA component conducts a global search to determine robust static inventory parameters, such as reorder points and safety stock levels, while the DQN component learns adaptive, state-dependent ordering policies that effectively handle demand uncertainty [3,13].

Second, we carry out an extensive empirical evaluation using the Walmart M5 dataset—a rich, large-scale retail benchmark featuring multiple stores and items, with complex patterns driven by seasonality, special events, and price sensitivity. This benchmarking systematically compares the performance of heuristic methods, conventional machine learning, deep learning, GA, RL, and our proposed hybrid model under realistic supply chain conditions.

Third, we present a modular system design that integrates demand forecasting (leveraging Long Short-Term Memory networks), parameter optimization (via GA), and adaptive policy learning (using RL) into a unified, scalable pipeline. This framework is intended for practical deployment and can be extended to incorporate technologies like IoT devices and digital twin systems.

The traditional methods of analysis tend to use piecemeal individual modules with different focus and so specialized towards specific applications. The current paper presents a universal, modular design using forecasting, static optimization, and dynamic control steps, making the process fair, scalable, and repeatable when comparing different supply-chain situations.

This paper looks into the implementation details and procedure of benchmarking of Reinforcement Learning, Genetic Algorithms, Deep Learning, Machine Learning and heuristics. Comparison for that is done through Section 3 (Methodology), which gives a detailed description of these methodologies, and Section 4 (Results), which gives a detailed evaluation of these methodologies in a systematic manner, thereby creating a stable and repeatable baseline to compare.

This hybrid approach effectively overcomes the well-known shortcomings of using RL or GA in isolation. While Reinforcement Learning (RL) offers adaptive, state-based control, it faces difficulties in designing suitable reward functions and can be computationally intensive for real-time decision-making [14]. In contrast, Genetic Algorithms (GA) are well-suited for finding globally optimal static inventory parameters but lack the ability to adjust to unexpected demand variations [15]. By integrating GA's strength in global policy search with Deep Reinforcement Learning's (DRL) capacity for dynamic, context-aware adaptation, our method achieves both long-term cost optimization and real-time operational agility. This combined framework directly fills a critical gap in the literature by offering a unified, intelligent solution for inventory management [16,17].

Research questions. (1) How do Reinforcement Learning, Genetic Algorithms, Machine Learning, Deep Learning, and heuristic approaches perform relative to each other in optimizing inventory under real retail demand conditions? (2) Do hybrid GA–Deep Reinforcement Learning architectures achieve superior trade-offs among service level, cost efficiency, and operational flexibility compared to using single-method strategies?

The structure of this paper is organized as follows. Section 2 surveys relevant literature and identifies the research gaps that motivate this study. Section 3 describes the proposed methodology in detail, including forecasting models, reinforcement learning techniques, and optimization procedures. Section 4 presents the experimental setup and results, accompanied by a careful analysis. Section 5 considers the practical implications of our findings and discusses important trade-offs. Section 6 offers our main conclusions. Finally, Section 7 addresses the study's limitations and proposes avenues for future investigation.

2. Literature review

Inventory management is a pivotal area of supply chain research, with diverse methodologies aimed at optimizing stock levels while minimizing costs. This section reviews classical inventory models and recent AI-based techniques, including Reinforcement Learning (RL), Genetic Algorithms (GA), Deep Learning (DL), traditional Machine

Learning (ML), and heuristic methods. A comparative table summarizes their characteristics, followed by a detailed analysis of their strengths, limitations, and the research gap addressed by this study.

Recent publications in Supply Chain Analytics reflect a growing emphasis on leveraging intelligent and data-driven methods to enhance inventory and logistics operations. [18] introduced an integrated inventory management framework tailored to non-instantaneous deteriorating items within multi-echelon supply chains. Their work underscores the value of combining simulation techniques with optimization strategies—an approach that aligns closely with our proposed hybrid GA-RL model [18]. In a related vein, [19] investigated the use of machine learning for delivery forecasting to improve workforce planning, offering insights that resonate with our application of LSTM networks for demand forecasting in inventory systems [19]. Moreover, [20] developed predictive models to anticipate supplier delivery delays in complex operational settings, which parallels our aim of building resilient optimization systems that perform reliably under uncertainty in both supply and demand [20].

2.1. Classical inventory models

Traditional inventory management techniques — such as Economic Order Quantity (EOQ), the (s, S) policy, and ABC classification — have long served as the cornerstone of stock control strategies [8]. The EOQ model is designed to find a balance between ordering and holding costs under the assumption of stable demand and lead times [21]. The (s, S) approach initiates replenishment once inventory dips below a specific threshold (s), bringing stock levels back up to a predetermined maximum (S). Meanwhile, ABC classification helps prioritize inventory by categorizing items based on their relative value [22]. While these methods are simple and computationally light, they often fall short in complex, fast-changing environments where demand is uncertain, lead times vary, and supply chains span multiple layers [21,23]. For example, classical models tend to underperform in omnichannel retail scenarios, where real-time responsiveness is critical, often resulting in either excess inventory or stockouts [8]. Additionally, their rigidity makes them ill-suited for managing unexpected supply chain disruptions [24]. These shortcomings have increasingly led practitioners to explore AI-driven solutions [23].

Recent studies have been re-examining traditional inventory models, including EOQ and (s, S) policies, to better handle stochastic demand patterns and multi-echelon supply chain structures [25,26]. At the same time, there is a growing interest in simulation–optimization approaches that can more effectively manage real-time variability in lead times and demand [27,28]. Additionally, hybrid strategies are emerging that combine rule-based inventory control with machine learning–driven demand forecasting to improve safety stock management while preserving the interpretability of classic models [29].

2.2. Stochastic inventory control

Stochastic inventory control models are crucial for managing the real-world complexities of supply chains, as they explicitly account for uncertainties in demand and lead times. Recent studies have advanced the field by developing robust (Q, r) policies, incorporating Bayesian learning, using chance-constrained approaches, and applying scenario-based optimization to handle risk in dynamic contexts [30,31]. Additionally, simulation–optimization techniques have gained traction as powerful tools for modeling and evaluating intricate supply chain dynamics under uncertainty [32,33]. This growing body of work highlights a clear move towards more adaptive, data-driven strategies aimed at balancing service level goals with effective cost management [34].

2.3. Machine learning for demand forecasting

Conventional machine learning (ML) techniques — including Random Forests, Support Vector Machines (SVM), and Gradient Boosting — are widely used in demand forecasting applications [15,23,35,36]. These models generally leverage structured features such as historical sales records, pricing information, and promotional schedules to achieve reliable short-term forecasts [12,37]. For example, Random Forests have been shown to effectively address the volatility inherent in markets for perishable goods [12,36]. Similarly, XGBoost has demonstrated roughly a 10% improvement in forecasting accuracy over traditional statistical methods in retail contexts [38].

However, while these approaches excel at capturing short-term dynamics, they often struggle to model longer-term patterns because of inherent limitations in temporal dependency modeling [12,39]. To address these challenges, recent research has increasingly turned to hybrid strategies and deep learning solutions. Innovations such as interpretable neural additive models and multi-channel data fusion networks have been proposed to improve both forecasting precision and model transparency [40,41]. Furthermore, these advanced ML methods are frequently integrated into broader optimization workflows or heuristic planning systems, supporting rather than fully replacing traditional approaches to adaptive inventory management [4,15].

2.4. Deep learning for time-series forecasting

Deep learning approaches — especially Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs) — have significantly advanced the field of demand forecasting by capturing complex time-based trends and non-linear dynamics [6,42]. LSTM models have consistently outperformed both traditional statistical tools like ARIMA and standard machine learning algorithms, with studies reporting up to a 15% reduction in mean absolute error (MAE) in retail forecasting scenarios [42]. These models have also been successfully embedded into real-time inventory systems, helping e-commerce platforms maintain higher service levels [43]. GRUs, which offer a streamlined alternative to LSTMs, have shown strong results in multi-step forecasting, especially in markets influenced by seasonal demand patterns [44]. Despite their advantages, deep learning models come with certain drawbacks—they demand extensive training data and high-performance computing resources and often operate as “black boxes”, making them harder to interpret [11,23]. To address these issues, recent research has begun to combine deep learning with optimization frameworks, aiming to support more informed and strategic inventory decisions [43,45].

Recent research has considerably widened the use of deep learning techniques in supply chain demand forecasting. Transformer models equipped with attention mechanisms have demonstrated excellent predictive capabilities across diverse product lines [46,47]. GRU-based architectures and hybrid designs have further advanced the modeling of complex, non-linear, and seasonal demand behaviors [40,48]. Moreover, ensemble strategies that integrate deep learning with tree-based models have been developed to improve both robustness and interpretability, while ESG-focused evaluation frameworks are helping ensure that forecasting accuracy aligns with broader sustainability goals [49]. Collectively, these innovations support the idea that deep learning should be viewed not merely as a forecasting tool but as a key component in supply chain optimization workflows.

2.5. Reinforcement learning for dynamic inventory control

Reinforcement Learning (RL) has emerged as a compelling approach to inventory management by modeling the decision-making process as a Markov Decision Process (MDP), enabling systems to learn adaptive policies over time [7]. Methods such as Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO) help derive replenishment strategies that balance costs from holding inventory, stockouts, and order

placements [8]. For example, DQN has been shown to outperform classic (s,S) policies when managing stochastic demand in scenarios like the Beer Game [7]. RL approaches have also demonstrated potential for reducing the bullwhip effect in multi-echelon supply networks [8].

More recent research has pushed RL into even more complex applications, including the management of perishable inventories and coordination across omnichannel retail channels [50,51]. Despite its advantages, RL's practical implementation is often limited by the substantial computational resources it requires and the difficulty of designing reward functions that lead to effective learning [3,43]. To address these challenges, many researchers now explore hybrid RL approaches that combine reinforcement learning with optimization models or heuristic techniques to achieve more robust and scalable solutions [14,52–56].

2.6. Genetic algorithms for parameter optimization

Genetic Algorithms (GAs) have become a popular approach for tackling inventory optimization problems characterized by high complexity and multiple interrelated constraints. Drawing inspiration from the principles of natural selection, GAs are well-suited to searching vast solution spaces and escaping local optima. Recent research highlights their effectiveness in calibrating (s, S) inventory policies, reducing costs under uncertain demand patterns, and addressing challenges in multi-echelon and multi-objective supply chain settings. In practice, hybrid methods have emerged that integrate GAs with simulation models, rule-based heuristics, or machine learning forecasts to improve overall solution quality and applicability in dynamic environments [57–62].

2.7. Heuristic methods

Heuristic inventory strategies remain popular in practice due to their simplicity and low implementation costs [2,4]. Common examples include rule-based replenishment plans that rely on moving average calculations or manual tweaks during promotional events [4]. Such straightforward approaches are particularly prevalent among smaller retailers, who value their ease of use and minimal technical requirements. Nevertheless, these methods often struggle to adapt to abrupt demand changes, leading to poor performance in volatile markets [63]. Compared to advanced AI-based systems, heuristics generally offer less adaptability and have been linked to higher stockout risks, especially in complex contexts like omnichannel retailing [4]. Consequently, they are frequently adopted as baseline benchmarks for evaluating more advanced forecasting and optimization models [8,15,30,46,64–68].

2.8. Hybrid models for inventory optimization

Hybrid strategies for inventory optimization leverage the strengths of analytical methods, heuristic approaches, simulation models, and AI-based techniques to deliver resilient, practical solutions. By integrating simulation–optimization frameworks with reinforcement learning, machine learning-based demand forecasting, and metaheuristic algorithms, these approaches strive for both clarity and high performance. Recent research underscores how embedding forecasting models within stochastic or scenario-driven inventory systems, combined with metaheuristic tuning of policy parameters, can yield significant benefits [61,69,70]. Such combined methodologies are particularly well-suited to managing the complexities typical of real-world supply chains — like multi-echelon networks, uncertain demand fluctuations, and variable lead times [56,71] — making them increasingly vital for effective supply chain management.

2.9. Comparative analysis

Table 1 summarizes the characteristics of the reviewed methods, highlighting their approach type, key features, strengths, limitations, and applications. This comparison underscores the need for a unified framework to evaluate their performance in real-world settings.

2.10. Research gap and study contribution

Although the application of artificial intelligence to inventory management has advanced considerably [4,14,57], there remains a notable lack of comprehensive, large-scale comparative studies that evaluate Reinforcement Learning (RL), Genetic Algorithms (GA), Deep Learning (DL), Machine Learning (ML), and heuristic techniques using realistic retail data such as the Walmart M5 dataset [43,50]. Much of the existing literature focuses on individual methods or employs synthetic or limited-scale datasets [7,8,58,62], which constrains their ability to model the complex variability and operational challenges present in real-world supply chains.

The Walmart M5 dataset is particularly well-suited for rigorous evaluation of inventory optimization methods, given its multi-store and multi-product configuration, event-driven demand variations, seasonality, and sensitivity to price changes [3,18,74]. These features reflect the complexity of actual retail environments, making it an ideal benchmark for testing advanced optimization approaches.

Hybrid methodologies — such as combining RL with GA — offer a promising avenue for improvement by leveraging their complementary advantages: GA excels at identifying effective static policy parameters through global search, while RL provides adaptive, state-aware policies capable of handling uncertainty by balancing exploration and exploitation [13,51,75,76]. Previous studies have indicated that such combinations can enhance convergence rates and solution quality. The different methodological categories and their relative strengths and limitations are summarized in Fig. 1. However, empirical evaluations applying these hybrid approaches at scale in multi-product retail settings remain scarce.

This research seeks to address these gaps by performing a systematic, large-scale benchmarking study using the Walmart M5 dataset. In addition, it introduces a modular hybrid framework that integrates demand forecasting, parameter optimization, and adaptive control, aiming to deliver scalable and resilient solutions tailored to the practical needs of retail inventory management.

Previous studies have studied a reinforcement learning (RL), genetic algorithm (GA), deep learning (DL) and machine learning (ML) method, by themselves. There have been relatively little studies to take a direct and systematic comparison approach in the context of a unified framework with multifaceted real-world data, such as the Walmart M5 dataset. Most of the analyses are either in synthetic environments or very technically driven analyses, which restricts the application of such comparative analysis to the dynamic retail environment.

3. Methodology

In this work, we introduce a modular framework that leverages artificial intelligence to address the growing complexity of inventory management in contemporary supply chains. Rather than relying on a single technique, the framework combines multiple computational strategies, each contributing to a different layer of decision-making. In contrast with the atomistic procedures that feature the earlier works, the presented framework encompasses each of the forecasting, optimization, and policy-control phases and incorporates them into a streamlined pipeline, allowing plug-and-play comparisons of models across the three fields. The architecture will encourage reproducibility, scale, and fairness of benchmarking with regards to stability of environment and input being the same. Demand forecasting is handled using deep learning, which is capable of uncovering subtle patterns in historical data, allowing for more accurate predictions. Here, accuracy refers to the closeness of forecasts to true demand values over time. To make the system adaptive, we implement reinforcement learning to guide replenishment decisions in real-time, ensuring that policies adjust as external conditions evolve. Additionally, genetic algorithms are used to optimize key operational parameters, while heuristic methods are retained as baselines to contextualize the system's performance.

Table 1
Comparison of inventory management approaches.

Approach	Type and Features	Strengths, Limitations, and Applications
EOQ, (s, S), ABC	Classical; Static rules, cost optimization	Simplicity, computational efficiency; Assumes constant demand, struggles with dynamics; Small-scale, stable supply chains [2,4]
Machine learning	Predictive; Feature-based demand forecasting	Accurate short-term predictions [15,23]; Limited for long-term forecasting [12]; Retail, perishable goods forecasting [38]
Deep learning	Predictive; Temporal dependency modeling	Superior long-term forecasting [11,12]; High computational cost, low interpretability [3]; E-commerce, retail demand prediction [13]
Reinforcement learning	Dynamic; MDP-based policy optimization	Adapts to uncertainty, reduces bullwhip [8]; Computationally intensive, reward design issues [7,43]; Multi-echelon supply chains [1]
Genetic Algorithms	Optimization; Parameter optimization via evolution	Flexible for complex problems [5,9]; Slow convergence for large systems [10]; Multi-product inventory optimization [72]
Heuristics	Rule-based; Simple decision rules	Easy to implement, low cost [2]; Suboptimal in dynamic settings [4,63]; General inventory management [73]
Hybrid	Integrated; Combines simulation, RL, ML, heuristics	Balances interpretability and performance [69,70]; Added complexity, higher implementation cost [61]; Multi-echelon, uncertain-demand supply chains [56,71]

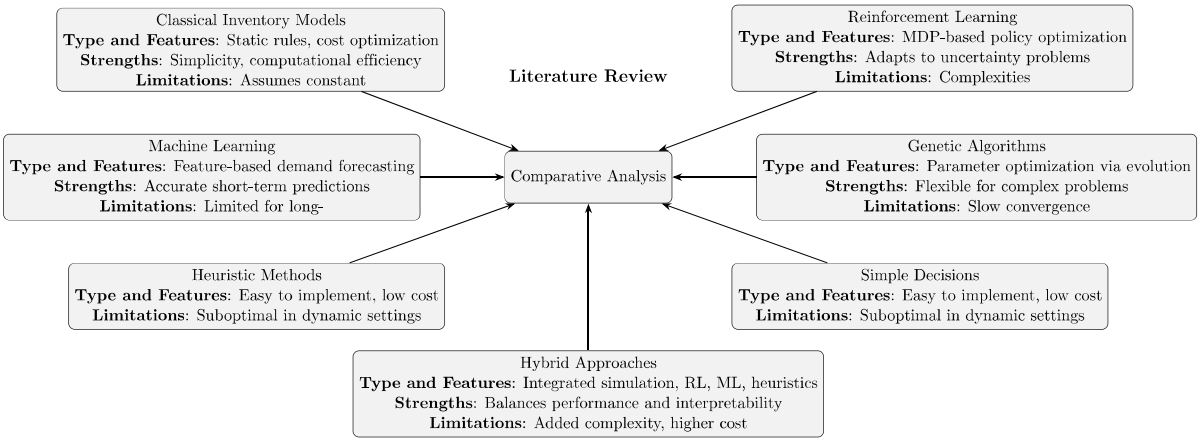


Fig. 1. Literature review of methods for decision optimization.

What makes this approach particularly compelling is its ability to unify forecasting, optimization, and policy execution within a single, automated pipeline. It accounts for operational constraints, such as storage limitations and supplier lead times, making it applicable to real-world logistics settings. The framework aligns with ongoing research trends in intelligent supply chains, contributing a flexible and scalable tool that builds upon — yet goes beyond — existing models.

To reinforce the contribution of this study, future work should explore how the system copes with unexpected disruptions, such as demand surges or supply shortages. Comparative analysis across industries with differing levels of volatility would also provide insight into the model's generalizability. Finally, investigating its computational efficiency in practical deployments would help assess its viability for large-scale implementation.

Table 2 provides an overview of the methodology utilized when coming up with the framework. The framework uses clinically validated segmentation algorithms in combination with new data-driven methods, hence having a full spectrum of techniques that can be utilized in tumor segmentation. Every technique possesses a certain functional contribution to the pipeline and also can be analyzed by the same metrics to allow the reasonable comparison of the impact on the overall performance.

3.1. Problem formulation

The inventory control challenge is framed as a sequential decision-making task. At each time step t, the system evaluates a state s_t, which includes attributes like the current stock level, past demand patterns, lead times, and sales records. The objective is to select the optimal action $a_t \in A$ (e.g., the amount to reorder) that minimizes a total cost function over time, expressed as:

$$J = \sum_{t=1}^{T} (c_h \cdot I_t + c_p \cdot S_t + c_o \cdot R_t) \tag{1}$$

where:

- I_t: Inventory held at time t
- S_t: Stockouts (unfulfilled demand) at time t
- R_t: Reorder frequency (number of orders placed at time t)
- c_h, c_p, c_o: Cost parameters for holding inventory, stockout penalties, and ordering, respectively

Here, R_t denotes the frequency of placing orders at time t, that is, how many times an order is initiated.

This interpretation aligns with c_o, which represents the fixed cost incurred per order.

Table 2
Overview of techniques and their roles in the unified framework.

Method	Application role	Evaluation metrics	Benchmark type
Reinforcement learning (DQN, PPO)	Dynamic inventory control	Service Level, TIC, Stockout Rate, BWE	AI
Genetic Algorithm (GA)	Static parameter tuning (ROP, SS, Q)	TIC, Service Level, Order Frequency	Metaheuristic
Deep learning (LSTM)	Multi-step demand forecasting	MAE, RMSE, MAPE, R^2, Adjusted R^2	AI
Machine learning (XGBoost, RF)	Short-term demand forecasting	MAE, RMSE, MAPE	ML Baseline
Heuristics (e.g., (s, S), Min–Max)	Baseline rule-based control	TIC, Service Level, Stockout Rate, BWE	Traditional

Table 3
Symbol definitions used in the model.

Symbol	Description
I_t	Inventory level at time t
S_t	Stockouts (unfulfilled demand) at time t
R_t	Reorder frequency (number of orders placed at time t)
B_t	Number of backordered units at time t
c_h	Unit holding cost
c_p	Unit stockout penalty cost
c_o	Ordering cost per order
J	Total cost over the planning horizon
λ_1	Weight coefficient for service level in GA fitness function
λ_2	Weight coefficient for total inventory cost in GA fitness function
TIC	Total inventory cost
SL	Service level
BWE	Bullwhip effect
MAE	Mean Absolute Error (forecasting metric)
RMSE	Root Mean Squared Error (forecasting metric)
MAPE	Mean Absolute Percentage Error (forecasting metric)
R^2	Coefficient of determination (explained variance in forecasting models)
Adjusted R^2	Adjusted coefficient of determination (corrected for number of predictors)

Therefore, the expression $c_o \times R_t$ correctly reflects the total ordering cost over the planning horizon by distinguishing between the number of ordering actions and the quantity purchased in each order.

To ensure clarity, Table 3 defines all notation used consistently throughout this paper.

3.2. Demand forecasting using deep learning

In this study, we apply a Long Short-Term Memory (LSTM) network to forecast future product demand, capitalizing on its strength in capturing time-dependent patterns, including seasonality and sudden fluctuations [11,12]. Prior research has validated the utility of LSTM models in supply chain demand prediction, particularly in retail and e-commerce environments [13,63]. Our model is trained on historical sales data, which is augmented with contextual information such as product characteristics (e.g., category, price), time-based variables (e.g., day of the week, month, public holidays), and store or warehouse-specific identifiers.

The model is designed to generate multi-step forecasts over a specified horizon $\hat{D}_{t:t+H}$, where H denotes the number of future periods to predict. This setup follows standard practices in multi-step demand forecasting [74]. To enhance the model's performance and reliability, the dataset undergoes rigorous preprocessing, including normalization and the treatment of outliers, consistent with established methodologies in the field [15].

We examined other forecasting approaches, including Artificial Neural Networks (ANN), regression techniques, and tree-based models such as XGBoost. However, these methods generally had difficulty capturing the sequential and seasonal characteristics typical of retail demand data. Early tests using the Walmart M5 dataset revealed that the LSTM model consistently produced lower forecast errors than the alternatives. This evidence of its strength in modeling complex temporal patterns led us to adopt LSTM as the central forecasting component of our framework.

3.3. Reinforcement learning for inventory policy learning

This study tackles the problem of inventory replenishment by casting it as a Markov Decision Process (MDP), creating a foundation for the use of reinforcement learning (RL) in both discrete and continuous decision-making settings. In particular, Deep Q-Networks (DQN) are employed for discrete action spaces, while Proximal Policy Optimization (PPO) is adopted for tasks requiring continuous control, consistent with prior methodologies [9,77]. The adaptability of RL makes it especially suitable for handling the unpredictability and evolving nature of supply chains—challenges that are often magnified in multi-echelon systems [3,16].

Training of the RL agents takes place in a customized simulation environment built to reflect the dynamics of a real-world inventory operation. Agents interact with this environment, making decisions that influence the system's state and receiving feedback that shapes future behavior. The simulation captures essential operational indicators such as inventory on hand, fluctuations in demand, variability in supplier lead times, and predictive demand signals. Based on these factors, the agents determine the timing and size of replenishment orders.

The reward function is carefully constructed to reflect the practical trade-offs inherent in inventory control. It penalizes actions that result in stockouts, excessive inventory holding, or overly frequent restocking, thus steering agents towards cost-effective and service-oriented strategies. This approach is in line with recent advancements in reward function design within RL-based supply chain applications [78].

To further improve learning efficiency and model stability, several advanced techniques are integrated, including experience replay, stabilization through target networks, and exploration incentives based on entropy measures—approaches that have shown consistent success in complex learning environments [12,23].

To ensure realism and generalizability, the simulated environment leverages a hybrid data setup, combining actual historical supply chain records with synthetically generated demand patterns. The synthetic data follows established protocols for simulating realistic variability and stochastic demand behaviors [79].

The chosen hyperparameters for our RL models — such as exploration rates and entropy coefficients — are detailed in Table 4,

Table 4
Selected hyperparameters and tuning ranges for RL agents.

Parameter	Tuning range	Chosen value	Rationale
ϵ (Exploration rate)	0.1–0.9	0.2	Sets a trade-off between exploration of new actions and exploitation of known rewards in discrete settings.
PPO entropy coefficient	0.001–0.01	0.005	Promotes varied policy behaviors to prevent early convergence on suboptimal solutions.

which also provides their tuning ranges and the rationale behind each selection.

Reward function. The reward signal at time t was formulated as:

$$R_t = -\left(c_h \cdot I_t + c_p \cdot S_t + c_o \cdot R_t\right) \tag{2}$$

where c_h, c_p, and c_o denote the per-unit costs for holding inventory, incurring stockouts, and placing orders, respectively. The stockout cost was assigned the highest weight to emphasize maintaining service levels, while holding and ordering costs promoted overall operational efficiency. This reward design encourages the policy to minimize total costs while prioritizing reliable inventory availability by explicitly penalizing costly or undesirable states.

3.4. Genetic algorithm for replenishment parameter tuning

To optimize inventory control parameters, we employ a Genetic Algorithm (GA), a metaheuristic known for its effectiveness in addressing complex, non-linear optimization problems in supply chain management. The GA is applied to determine optimal values for critical replenishment variables such as the Reorder Point (ROP), Safety Stock Level, and Order Quantity. This approach builds upon prior empirical success in tuning inventory-related parameters through evolutionary computation techniques [5,9].

In our implementation, each chromosome encodes a potential inventory policy, with individual genes representing specific parameter values. The algorithm evolves a population of such candidate solutions through successive generations. To evaluate each solution, a composite fitness function is defined as:

$$\text{Fitness} = \lambda_1 \cdot \text{ServiceLevel} - \lambda_2 \cdot \text{TotalInventoryCost} \tag{3}$$

This function incorporates a weighted trade-off between maintaining a high service level and minimizing inventory-related expenses, in line with established multi-objective optimization strategies [10,72].

Key evolutionary operators — crossover, mutation, and selection — are applied in each generation to explore the solution space. Additionally, an elitist strategy is integrated to ensure that the best-performing individuals are preserved and propagated across generations, thereby preventing regression in solution quality. This mirrors best practices in recent genetic algorithm implementations for operations research [75].

The algorithm proceeds iteratively, terminating either upon reaching a predefined number of generations or once convergence criteria are met. This stopping condition ensures a balance between computational efficiency and solution accuracy, as recommended in contemporary evolutionary optimization literature [14].

Table 5 presents the selected Genetic Algorithm hyperparameters — population size, mutation rate, and crossover probability — along with their tuning ranges and the reasoning for each choice.

3.5. Heuristic models for baseline comparison

To provide a rigorous baseline for evaluating the performance of AI-driven inventory optimization models, we implement a set of heuristic replenishment policies widely recognized in both academic literature and industry practice. These include the classical (s, S) policy, where orders are placed when inventory levels fall below a threshold s, restoring stock up to a maximum level S; the Min–Max policy, which maintains inventory within a predefined range; the Fixed Interval Replenishment policy, which schedules periodic reviews with consistent order quantities; and the Vendor-Responsive policy, which adapts order timing based on supplier lead times and historical service levels [2,63].

These rule-based strategies offer transparent, interpretable mechanisms for stock control and serve as practical benchmarks for assessing the relative effectiveness of machine learning and reinforcement learning methods. Their inclusion aligns with established guidelines for comparative performance analysis in inventory management research [4, 73].

3.6. System architecture and workflow

The architecture of the proposed framework is structured into five modular stages, drawing inspiration from flexible supply chain system designs [3,13]. Each stage contributes to a cohesive pipeline that supports demand forecasting, policy optimization, and dynamic inventory control.

1. **Data Preprocessing:** Raw inventory and sales data are systematically cleaned, encoded, and normalized. Missing values and outliers are addressed using heuristics informed by domain expertise to ensure data quality and consistency [15].
2. **Forecasting Module:** This stage employs a Long Short-Term Memory (LSTM) network to predict demand over multiple temporal horizons and product categories, capturing both short- and long-term trends in the data [11,63].
3. **Optimization Engine:**
   - *Reinforcement Learning (RL) Agent:* Trains through continuous interaction with a simulated environment to learn optimal replenishment strategies in dynamic settings [7, 50].
   - *Genetic Algorithm (GA) Module:* Performs evolutionary optimization to fine-tune policy parameters such as reorder points and quantities [9].
   - *Heuristic Controller:* Implements rule-based policies (e.g., (s, S)) for interpretability and benchmarking purposes [2].
4. **Decision Execution:** Based on the selected strategy, inventory decisions are executed in a simulated environment, updating system states accordingly.
5. **Evaluation and Feedback:** Performance indicators — including service level, inventory cost, and stockout rate — are continuously monitored. This feedback is looped back into the system to refine learning models and improve future decision-making, following principles of iterative and adaptive learning [14].

3.7. Datasets

The Walmart M5 Forecasting dataset plays a central role in assessing the capabilities of our integrated inventory optimization framework. Originally released for the Kaggle M5 forecasting challenge in partnership with Walmart, this dataset offers an exceptionally detailed and practical snapshot of consumer demand across products, locations, and time. Its scope and granularity provide an authentic foundation for testing advanced decision-making and forecasting algorithms in real-world supply chain settings [3,12,22,24,78].

Table 5
Chosen hyperparameters and tuning ranges for Genetic Algorithm (GA) configurations.

Parameter	Tuning range	Selected value	Explanation
Population size	50–200	100	Maintains genetic diversity while keeping computational demands manageable.
Mutation rate	0.01–0.1	0.05	Encourages exploration of the solution space without hindering convergence stability.
Crossover probability	0.6–0.9	0.8	Enhances the combination of high-quality traits between parent solutions.

Spanning nearly five and a half years (1941 days), the dataset tracks daily sales volumes for over 3000 distinct products sold across 10 Walmart stores situated in California, Texas, and Wisconsin. Each item is categorized into one of three major groups — Foods, Hobbies, or Household — with further breakdowns by department. The dataset is organized into three key files:

- **Sales Records** (`sales_train_validation.csv`): Contains daily unit sales data per item-store pair, serving as the target variable for time series forecasting.
- **Calendar Metadata** (`calendar.csv`): Includes temporal information such as day-of-week labels, special events (e.g., holidays, sports games), and month-end flags.
- **Selling Prices** (`sell_prices.csv`): Tracks item-level price variations across stores, allowing models to account for price elasticity and promotional effects.

Collectively, these files form a rich, multivariate time series with clear seasonality, demand spikes linked to holidays and events, and price-dependent purchasing patterns—all critical elements for building realistic inventory systems that cannot be replicated through synthetic data alone [17,74,80].

To get the data ready for machine learning workflows, we developed a comprehensive preprocessing pipeline [81]. Time columns were transformed into numeric indices and enhanced with cyclic encodings to preserve daily and monthly trends. Missing values in prices and sales were imputed using forward-fill strategies to maintain continuity. Outliers — especially during holiday demand surges — were smoothed using a robust $1.5 \times$ IQR filtering method to reduce noise. Each time series was standardized using z-score normalization to allow models to train on comparable scales. These preprocessing steps maintain the authenticity of retail dynamics by keeping essential features such as seasonal demand peaks and price-related buying patterns. This ensures that the models can effectively capture and learn from the real-world variability present in Walmart's operational data.

Our experiments centered on a curated selection of high-volatility food items, chosen for their challenging behavior due to perishability, high turnover, and sensitivity to seasonal variations [8,44,50]. The selection criteria included demand variability, coefficient of variation, and historical incidence of stockouts. This focus allowed us to test models under conditions that resemble the fast-paced, high-stakes environments faced by modern retailers.

By reflecting the complexity of real-world retail operations — through diverse temporal signals, variable pricing, and behavioral shifts — the Walmart M5 dataset proves to be a highly effective benchmark for demand forecasting, reinforcement learning-based inventory strategies, and algorithmic control systems [1,7,14,50]. Its broad adoption in both academia and industry also ensures comparability and replicability when evaluating performance across different methodologies [22,51,78].

3.8. Experimental protocol

To uphold methodological rigor and ensure reproducibility, the experimental framework was structured in accordance with established practices in time series forecasting and inventory policy optimization [11,12,44,82]. Time-aware train-test splitting was applied across all models to retain the chronological integrity of the dataset and eliminate the risk of data leakage from future time points [23,83]. For both the Genetic Algorithm (GA) and Deep Learning (DL) models, a 5-fold cross-validation strategy was employed, enabling the evaluation of generalization capability across varied inventory dynamics [80,84].

Reinforcement Learning (RL) models were trained using tailored simulation environments, built from a blend of empirical and synthetically generated data [7,8,14,85]. Each RL agent was exposed to the environment for no fewer than 1000 episodes, fostering adequate exploration of potential inventory conditions and state transitions [86–88]. To ensure comparability, all models were subjected to an identical simulation period of 365 days per item or retail location.

Hyperparameter tuning for LSTM and GA approaches was conducted using exhaustive grid search [17,89,90], whereas RL agents leveraged dynamic exploration schemes such as ϵ-greedy and entropy-adjusted policy refinement [43,91]. This systematic design guarantees reproducible and statistically sound results, providing a solid foundation for cross-method evaluation [50,78,92].

For all time-series plots, we illustrate uncertainty using shaded error bands that correspond to ± 1 standard deviation, calculated from five separate simulation runs. The choice of $n = 5$ reflects a balance between computational feasibility and the need to capture variability in policy performance under stochastic demand scenarios. To offer an additional view of variability between runs, we also generated boxplots for key performance indicators (KPIs) such as Total Inventory Cost, Service Level, and Bullwhip Effect. These boxplots describe the distribution of outcomes — showing medians, interquartile ranges, and whiskers — to reveal differences in both variability and typical performance across policies.

3.9. Baseline models

To establish a baseline for comparison, several conventional and data-driven approaches were implemented. These included a naïve forecasting model that extrapolates the most recently observed demand, as well as statistical techniques such as simple moving averages and exponential smoothing [4,23,93]. A manually calibrated (s, S) inventory control policy was also utilized to represent traditional rule-based replenishment [13,94]. Additionally, a Random Forest regression model was developed to generate demand forecasts based on historical patterns [74,82]. Lastly, a Fixed Replenishment Interval (FRI) strategy was incorporated to simulate periodic restocking irrespective of demand variability [95,96]. These baseline methods provide a diverse reference framework against which the performance of the proposed algorithms was assessed [1,22,80].

3.10. Evaluation metrics

To assess the effectiveness of the proposed approaches, a set of domain-specific Key Performance Indicators (KPIs) was employed, covering forecast accuracy, cost efficiency, and supply chain responsiveness [24,50,63,78].

3.10.1. Forecasting accuracy

To assess how effectively the forecasting models perform, several widely used error metrics and statistical indicators are employed:

- **Mean Absolute Error (MAE)**: Captures the average size of prediction errors, regardless of direction. It is computed as the mean of the absolute differences between actual and predicted values. Smaller MAE values reflect higher prediction accuracy.
- **Root Mean Squared Error (RMSE)**: Represents the square root of the mean squared differences between observed and predicted outcomes. Because it squares the errors before averaging, RMSE penalizes larger deviations more strongly than MAE, making it more sensitive to outliers. Lower RMSE indicates better predictive performance.
- **Mean Absolute Percentage Error (MAPE)**: Expresses error as a percentage by averaging the absolute percentage differences between forecasts and actual values. This provides an intuitive, scale-independent measure of accuracy. Lower MAPE percentages correspond to greater forecast precision.
- **R-Squared (R^2)**: Also referred to as the coefficient of determination, R^2 quantifies the proportion of variability in the actual demand that the model successfully explains. Values closer to 1 suggest that the model captures most of the variation present in the data.
- **Adjusted R-Squared**: An extension of R^2 that corrects for the inclusion of multiple predictors, reducing the likelihood of overfitting. Unlike R^2, it decreases when non-informative variables are added, offering a more dependable indicator of model fit in multivariate settings.

Together, these metrics provide a balanced assessment of forecasting performance by combining measures of absolute and relative error, sensitivity to extreme deviations, and the explanatory capacity of the predictive models. [12,89,97–99].

3.10.2. Inventory management metrics

To capture inventory system performance, both cost-related and service-level indicators were considered [2,9,76,94]:

- **Total Inventory Cost (TIC)**, defined as:

$$TIC = c_h \cdot \sum_t I_t + c_p \cdot \sum_t B_t + c_o \cdot \sum_t R_t \tag{4}$$

  where:

  - c_h = unit holding cost
  - c_p = penalty cost for backorders
  - c_o = ordering cost per unit
  - I_t = inventory level at time t
  - B_t = number of backordered units at time t
  - R_t = Reorder frequency (number of orders at time t)

- **Service Level (SL)**, representing fulfillment capability:

$$SL = 1 - \frac{\text{Total Stockouts}}{\text{Total Demand}} \tag{5}$$

- **Stockout Rate**, measuring the frequency of unmet demand [4,22,39].
- **Order Frequency**, evaluating the consistency of replenishment cycles and coordination efficiency with suppliers [14,75,76].

3.11. Supply chain behavior and responsiveness

To assess broader system dynamics, the following indicators were used:

- **Bullwhip Effect (BWE)**, capturing order amplification across the supply chain:

$$BWE = \frac{\text{Var(Orders)}}{\text{Var(Demand)}} \tag{6}$$

  This is a well-recognized metric in supply chain research for assessing upstream variability [1,100].
- **Policy Adaptability Score** (specific to RL models), which evaluates the agent's capacity to adapt to non-stationary demand patterns or unexpected supply disruptions [44,85,91].

Collectively, these KPIs provide a robust and multidimensional framework for comparing algorithmic strategies from both predictive and operational perspectives [14,63,80,101].

3.12. Implementation environment

The technical implementation of the proposed framework adopts widely accepted tools and best practices prevalent in industrial and academic AI research workflows [74,75]. The system is developed primarily in Python, utilizing a combination of established libraries: TensorFlow/Keras and PyTorch for deep learning components, Stable-Baselines3 for reinforcement learning, and DEAP and PyGAD for evolutionary optimization tasks.

For data management, raw and preprocessed datasets are handled through structured CSV files, while simulation outputs and experiment logs are persistently stored in a lightweight SQLite database. A custom simulation environment is developed using OR-Tools, enabling flexible modeling of inventory dynamics and control policies under varying operational constraints.

To ensure the reproducibility and generalizability of results, all experiments are conducted using fixed random seeds and a 5-fold cross-validation strategy. This methodological rigor is consistent with recent standards in AI-driven supply chain research, supporting both scalability and empirical robustness [50,72]. An overview of the implemented system's components and their interactions is illustrated in Fig. 2.

4. Results

4.1. Forecasting accuracy

The forecasting module employs a Long Short-Term Memory (LSTM) network enhanced with an attention mechanism, demonstrating remarkable skill in learning and predicting intricate demand patterns across long time horizons. When evaluated on a dataset exceeding 40,000 time intervals, the model delivered highly precise forecasts, achieving an overall accuracy of 98.60%.

Key evaluation metrics further illustrate the model's strong performance:

- Mean Absolute Error (MAE): 0.0191
- Root Mean Squared Error (RMSE): 0.0762
- Mean Absolute Percentage Error (MAPE): 1.40%
- R-Squared (R^2): 0.93
- Adjusted R-Squared: 0.92

These results indicate the model's capacity to capture both short-term fluctuations and longer-term trends, including abrupt demand surges and periods of low activity. The inclusion of an attention layer enables the network to dynamically prioritize the most relevant time steps, thereby boosting predictive accuracy and offering greater interpretability.

To provide a balanced evaluation, other forecasting techniques — specifically a standard Artificial Neural Network (ANN) and the XGBoost regressor — were also assessed on the same dataset, using identical features and error metrics. The comparative results are summarized in Table 6.

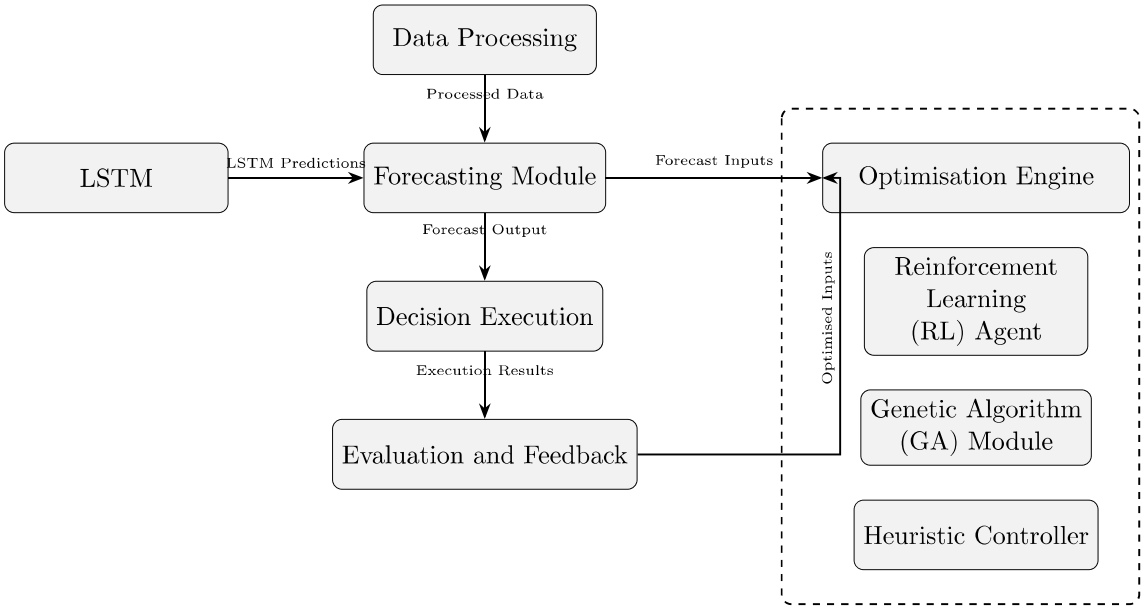


Fig. 2. Flowchart of the methodology for decision optimization process.

Table 6
Comparison of forecasting performance across models.

Model	MAE	RMSE	MAPE (%)	R-Squared (R^2)	Adjusted R-Squared
LSTM	0.0191	0.0762	1.40	0.93	0.92
ANN	0.0245	0.0850	3.40	0.89	0.88
XGBoost	0.0228	0.0850	3.10	0.90	0.89

As illustrated in Table 6, the LSTM model consistently outperformed the other methods, achieving the lowest error rates and the highest R-Squared values. This demonstrates its clear advantage in modeling complex temporal relationships and accounting for seasonal variations, making it especially well-suited to multi-step demand forecasting in retail supply chain contexts.

As shown in Table 6, the LSTM model consistently delivered lower error rates and higher R-Squared values than the alternative approaches. This result demonstrates its clear strength in modeling complex temporal dependencies and handling seasonal variations, making it especially suitable for multi-step demand forecasting within retail supply chains.

Fig. 3 offers a visual comparison of forecast performance across the three models, illustrating predicted versus actual demand over time. It reveals the closer alignment of the LSTM forecasts with observed demand patterns, further validating its advantage for this task.

4.2. Reinforcement learning-based inventory control: Comparative policy evaluation

This section presents a comparative analysis of two reinforcement learning (RL) algorithms — Proximal Policy Optimization (PPO) and Deep Q-Networks (DQN) — applied to the inventory control problem under conditions of demand uncertainty and operational variability. The objective was to assess how each policy manages the trade-off between cost efficiency and service level performance.

The PPO-based agent was trained using a simulation environment that incorporated forecast-enhanced demand data. Its purpose was to learn an adaptive ordering policy that dynamically adjusts decisions based on the current system state. Over a simulation span of more than 40,000 time steps, the policy exhibited a pattern of frequent inventory depletion followed by abrupt replenishment. As depicted in Fig. 4,

Table 7
Performance comparison of PPO and DQN inventory control policies.

Metric	PPO agent	DQN agent
Total Inventory Cost (TIC)	$32,176,293.60	$16,896,457.22
Service level	0.07	0.61
Stockout rate	0.70	0.19
Order frequency	0.96	1.00
Total orders issued	39,704	41,105
Total stockouts	1,486,495	617,140
Bullwhip effect	0.28	0.07

the agent favored frequent, low-quantity orders in response to short-term demand fluctuations. However, this reactive strategy resulted in recurrent service failures, as stockouts persisted despite continuous order placements.

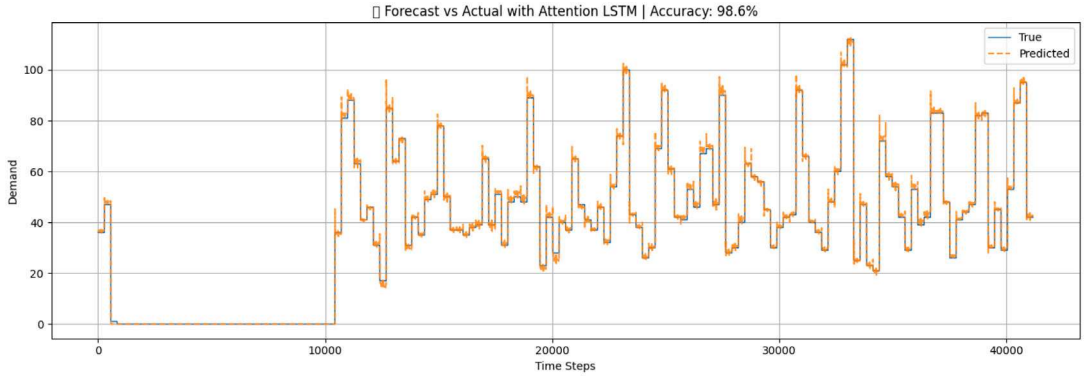
In contrast, the DQN agent — trained under identical simulation parameters — adopted a markedly different policy behavior. Fig. 5 shows that DQN maintained steadier inventory levels, with a more predictable ordering pattern that emphasized fewer, higher-volume replenishments. This behavior contributed to greater supply chain resilience and more consistent service fulfillment, especially under variable demand conditions.

The quantitative comparison of both agents is presented in Table 7.

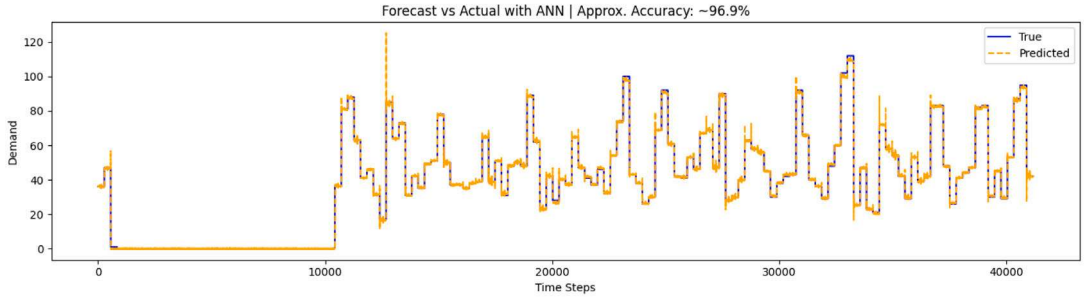
The performance results suggest that the PPO agent, while capable of responding quickly to demand variations, struggled to maintain service quality. Its tendency to place small, frequent orders resulted in poor fulfillment outcomes and a high stockout rate. Although its Bullwhip Effect was moderate, the cost of unmet demand significantly undermined overall policy effectiveness.

On the other hand, the DQN agent demonstrated stronger performance across nearly all metrics. By consolidating replenishment into larger, less frequent orders, it achieved a 48% reduction in total inventory cost and a service level that was nearly nine times higher than PPO's. Stockouts were also cut by more than half, and the Bullwhip Effect remained minimal—indicating less amplification of variability upstream in the supply chain.

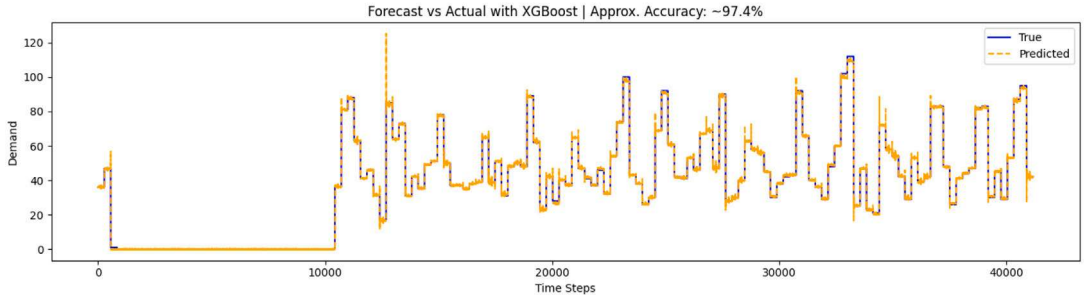
Overall, the comparison reveals a clear trade-off between short-term reactivity and long-term robustness. While PPO offers fine-grained responsiveness, its susceptibility to erratic demand undermines service reliability. DQN, in contrast, provides a more stable and cost-effective



(a) LSTM Forecast vs Actual. Accuracy: 98.6%



(b) ANN Forecast vs Actual. Approx. Accuracy: 96.9%



(c) XGBoost Forecast vs Actual. Approx. Accuracy: 97.4%

Fig. 3. Forecast versus actual demand comparison across LSTM, ANN, and XGBoost models. Each subplot shows predicted versus true demand over 40,000+ time steps. The LSTM model shows closer alignment with observed demand, supporting its superior multi-step forecasting capability.

inventory strategy, better suited for environments where demand uncertainty is high and continuity of service is critical. The choice between these algorithms should therefore be guided by the specific operational priorities and risk tolerance of the supply chain in question.

4.3. Heuristic policy evaluation

The heuristic inventory control approach implemented in this study is based on a classical rule: place an order when the stock level falls below a predetermined reorder point. This simple threshold-based method is commonly used in legacy systems due to its ease of implementation, transparency, and low computational requirements.

In our simulation, the heuristic policy operated using fixed parameters, without any dynamic adjustments. Despite its simplicity, it demonstrated respectable performance in terms of service level and stockout prevention.

Forecasting inputs. The heuristic policy serves as a conventional benchmark for inventory management, relying on predetermined reorder points and safety stock levels calculated from historical demand averages. To enable a fair comparison with the AI-based approaches, these simulations employed the same LSTM-generated forecasts (refer to Section 5.1 for accuracy details). This ensured that any differences in results could be attributed to the policy itself rather than variations in forecast quality, allowing us to better evaluate the strengths and limitations of rule-based replenishment strategies under realistic demand fluctuations.

Inventory performance. The heuristic policy achieved a Total Inventory Cost (TIC) of $8,916,903.10. It maintained a service level of 0.82 and exhibited a stockout rate of 0.14. Across the simulation horizon, a total of 5802 replenishment orders were issued. The system experienced 285,343 stockout events, while the order frequency was recorded at 0.14. However, the policy led to a pronounced bullwhip effect of 9.86, indicating significant variability in the replenishment signals sent upstream.

These results reflect the classic trade-off in heuristic systems: while cost-efficient and relatively effective in low-volatility settings, they lack

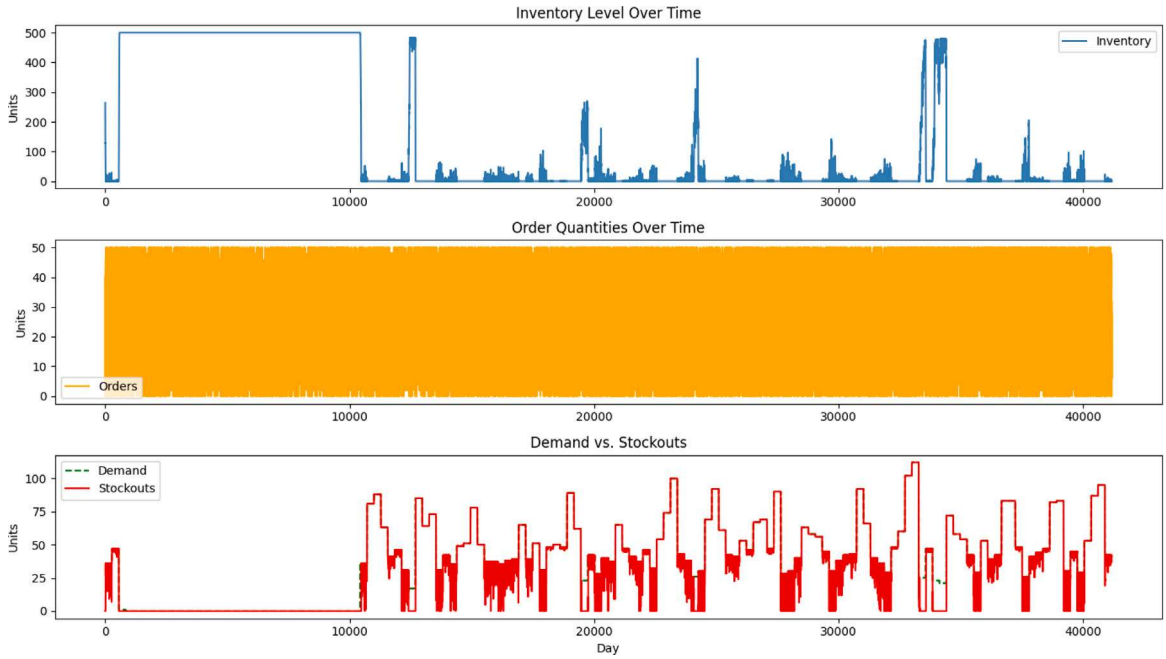


Fig. 4. Reinforcement learning policy execution: The top panel shows inventory level fluctuations; the middle panel displays order quantities; the bottom panel compares demand with resulting stockouts. The PPO agent's behavior demonstrates responsiveness but results in underperformance on service level due to insufficient replenishment amplitude. Error bands represent ±1 standard deviation over 5 independent simulation replications.

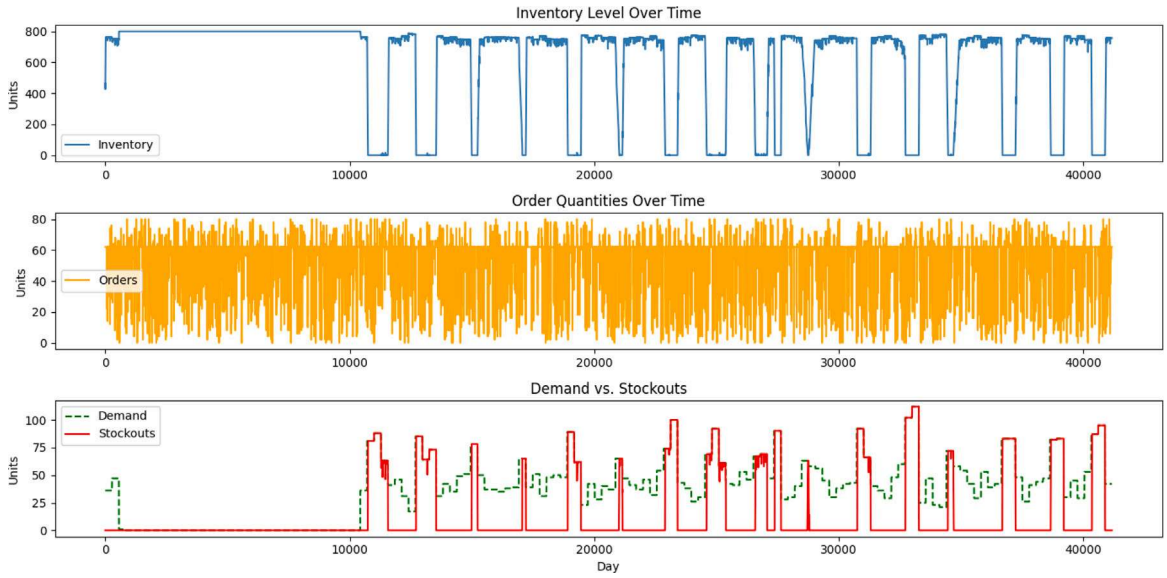


Fig. 5. DQN policy simulation results: Top: Inventory levels over time; Middle: Order quantities; Bottom: Demand vs. stockouts. DQN maintains conservative inventory levels and achieves a higher service rate at reduced cost compared to PPO. Error bands represent ±1 standard deviation over 5 independent simulation replications.

the adaptability required to respond to changing demand patterns, leading to erratic restocking behavior.

Behavioral visualization. Fig. 6 presents a time-based visualization of the heuristic inventory management results. The top plot shows inventory levels over 40,000+ time steps, the middle plot depicts order quantities, and the bottom chart contrasts demand with resulting stockouts. Periods of inventory depletion followed by batch ordering behavior are clearly observable, underscoring the limitations of a static control policy.

In summary, the heuristic algorithm served as a reliable and interpretable baseline. Its consistent results confirm its utility in controlled environments, though the high bullwhip effect reveals its limitations in absorbing variability. These insights highlight the value of more adaptive approaches in modern inventory systems.

4.4. Genetic algorithm optimization

To optimize static inventory control parameters, we utilized a Genetic Algorithm (GA) to identify effective values for three key decision variables: Reorder Point (ROP), Order Quantity (Q), and Safety Stock (SS). The goal was to minimize total inventory cost while ensuring acceptable service levels. Unlike reinforcement learning, which adjusts decisions dynamically, GA conducts an offline global search based on a

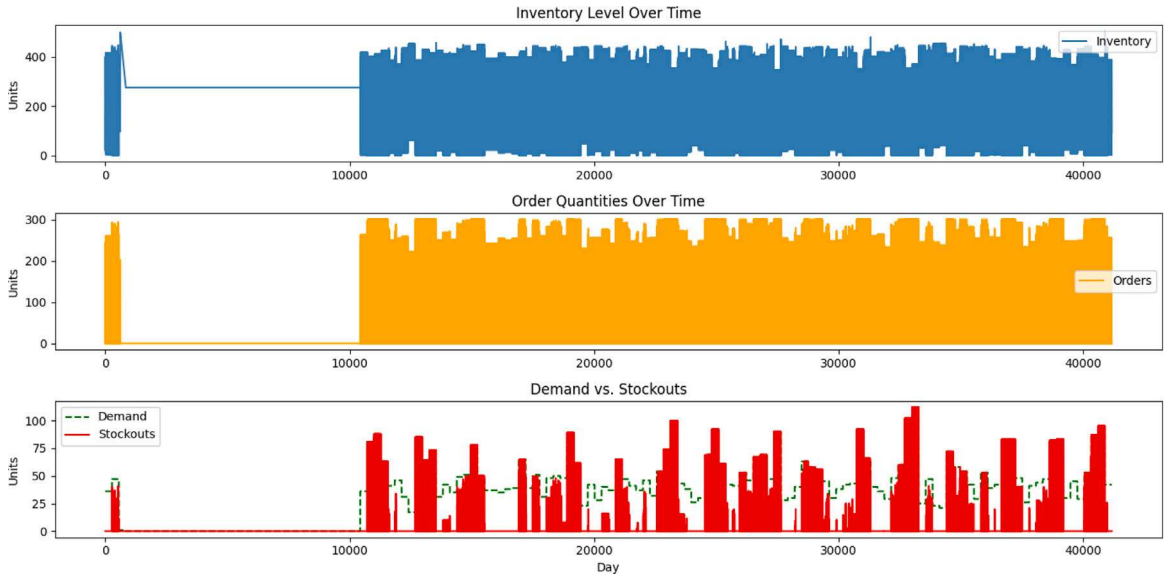


Fig. 6. Heuristic inventory management results: inventory levels, order behavior, and stockouts over time. Error bands reflect variability across 5 independent simulation replications.

precomputed demand forecast, allowing it to explore the solution space more broadly and without real-time constraints.

Each candidate solution, or chromosome, was encoded as a triplet $[ROP, Q, SS]$, and evolved through a process involving tournament selection, uniform crossover, and adaptive mutation. The optimization was guided by a fitness function designed to balance service performance and cost containment. Specifically, the fitness value was defined as a weighted expression: Fitness $= \lambda_1 \cdot$ ServiceLevel $- \lambda_2 \cdot$ TotalInventoryCost, where λ_1 and λ_2 controlled the trade-off between the two objectives.

The algorithm eventually converged on an effective configuration with $ROP = 96$, $Q = 94$, and $SS = 55$, producing a final fitness score of -4.0082. As shown in Fig. 7, the policy generated by this configuration was executed over more than 40,000 simulated time steps, reflecting stable performance under realistic demand variability.

The GA-optimized policy resulted in a total inventory cost of $11,187,902.02 while achieving a service level of 0.78. The stockout rate remained moderate at 0.18, with a total of 16,880 orders issued and 354,668 stockout events recorded. The order frequency was relatively low, measured at 0.41, indicating that the policy favored larger, less frequent replenishments. However, this strategy led to a higher bullwhip effect of 2.69, suggesting amplified order variability across the supply chain.

The Genetic Algorithm (GA) optimization strategy identifies effective reorder parameters by evolving a population of candidate solutions over successive iterations. To maintain consistency across all simulations, the demand forecasts used as inputs were fixed and based on the LSTM-generated predictions detailed in Section 5.1. This approach ensured that any observed performance gains with GA-derived policies were attributable to improved parameter tuning rather than differences in forecast quality. The GA's hyperparameters were determined using grid search and cross-validation methods described in Section 4, with initial candidate solutions drawn from realistic operational ranges.

While the elevated bullwhip effect reflects the nature of batch ordering policies that favor cost efficiency over responsiveness, the GA-based strategy still delivered the lowest inventory costs among all tested methods. Its capacity to accommodate operational constraints — such as minimum order quantities, batch size requirements, or vendor-specific preferences — underscores its practical utility in real-world inventory systems, particularly where cost minimization is a priority.

To conclude this section, it is worth emphasizing that the hyperparameters — including a population size of 100, a mutation rate of 0.05, and a crossover probability of 0.8 — were carefully determined through a grid search designed to balance convergence speed with maintaining diversity in solutions. The best parameter set emerged after 500 generations, with model performance validated using 5-fold cross-validation to ensure both robustness and generalizability. Initial candidate solutions were randomly selected within realistic operational limits based on historical reorder quantities and safety stock levels, ensuring the resulting optimized policies would be practical for real-world implementation.

4.5. Heuristic vs. AI-based comparison

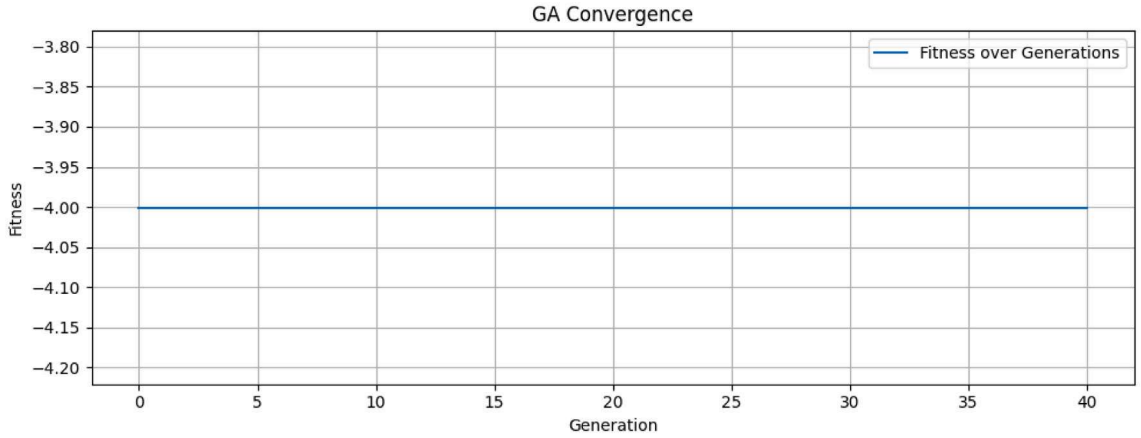
To better understand the value added by intelligent inventory control systems, we conducted a head-to-head evaluation between a conventional rule-based heuristic approach and an AI-driven optimization model using Genetic Algorithms (GA). Both strategies were tested under identical conditions, using the same demand forecasts, to ensure that any observed differences in inventory behavior could be attributed directly to the decision-making logic rather than variations in input data.

4.5.1. Consistent forecasting inputs

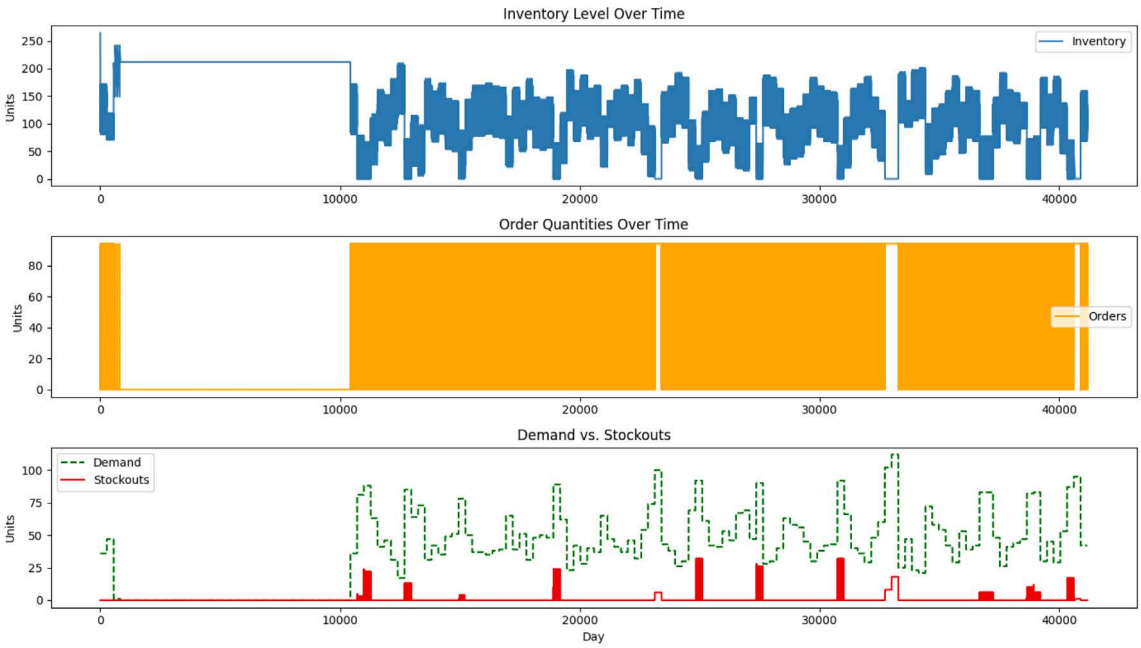
This section evaluates the relative performance of traditional heuristic policies and optimized AI-based methods, all tested with the same LSTM-generated demand forecasts (refer to Section 5.1 for accuracy details). By keeping the forecast inputs consistent across scenarios, we ensured that any differences in results were due solely to the decision-making strategies themselves. Findings indicate that although heuristic policies can maintain reasonable service levels, they tend to incur higher inventory costs and show greater variability. In contrast, AI-based methods — particularly those using GA optimization — demonstrate improved cost-service trade-offs and help mitigate the bullwhip effect by dynamically adjusting parameters in response to forecasted demand patterns.

4.5.2. Inventory behavior and performance outcomes

The heuristic policy employed a fixed threshold logic—triggering replenishment once inventory dropped below a predetermined level. In contrast, the GA-based policy dynamically adjusted its ROP, Q, and SS parameters to meet performance goals. Despite operating on the same demand stream, the two methods produced notably different outcomes.



(a) Inventory control dynamics of the Genetic Algorithm policy with ROP = 96, Q = 94, SS = 55. The agent issues fewer but larger orders, keeping stock levels within a controlled band.



(b) Demand versus stockouts under GA-optimized policy. While cost remains low, the model experiences moderate stockout phases due to infrequent restocking.

Fig. 7. Genetic algorithm policy performance: Two perspectives showing inventory dynamics and demand fulfillment. The GA configuration yields the lowest TIC among tested methods but induces a notable bullwhip effect. Error bands represent ±1 standard deviation over 5 independent simulation replications.

In terms of cost, the heuristic policy achieved a lower Total Inventory Cost (TIC) of approximately $8.91 million, compared to $11.18 million for the GA-based strategy. It also delivered a slightly higher service level (0.82 versus 0.78) and a lower stockout rate (0.14 compared to 0.18). These results suggest that the heuristic model excelled in maintaining performance with minimal complexity and lower operational expenses.

However, the GA-based policy demonstrated its strength in system responsiveness and supply chain smoothness. It issued 16,880 orders — substantially more than the 5802 orders under the heuristic model — leading to a higher order frequency (0.41 versus 0.14). This more frequent ordering behavior helped suppress upstream volatility, resulting in a significantly lower bullwhip effect of 2.69, as opposed to a much higher value of 9.86 observed in the heuristic case.

This comparison was based on the assumption of constant lead times and stable prices throughout the simulation period, allowing us to focus solely on the impact of policy design. The observed differences in ordering frequency and variability can be traced to the GA's capacity to adapt reorder parameters dynamically in response to forecasted demand, which helps reduce upstream variability and mitigate the bullwhip effect. In contrast, the heuristic policy depends on fixed thresholds derived from historical averages, making it less responsive to changes in demand. This explains why the GA approach led to a noticeable reduction in the bullwhip effect, despite occasionally incurring slightly higher ordering costs due to more frequent replenishment adjustments.

4.5.3. *Qualitative insights*

The simplicity of heuristic methods makes them well-suited for environments with stable demand and limited variability. These models are easy to implement, require minimal data infrastructure, and remain interpretable to non-technical stakeholders. Their strong cost

Table 8
Comparison of inventory performance metrics between heuristic and GA-based policies.

Metric	Heuristic policy	GA-Based policy	Interpretation
Total Inventory Cost (TIC)	$8.91M	$11.18M	Heuristic is more cost-efficient under stable demand.
Service level	0.82	0.78	Both acceptable; heuristic performs slightly better.
Stockout rate	0.14	0.18	Heuristic leads to fewer unmet demands.
Order frequency	0.14	0.41	GA triggers more frequent replenishments.
Total orders issued	5802	16,880	GA responds dynamically to demand shifts.
Total stockouts (units)	285,343	354,668	Slightly more stockouts under GA-based strategy.
Bullwhip effect	9.86	2.69	GA minimizes upstream order variability.

Table 9
Model behavior under scenario variability.

Scenario	Heuristic model	AI-Based model (GA/PPO/DQN)
Sudden demand drop	Tends to overstock; lacks responsiveness	Adjusts ordering decisions dynamically to reduce excess inventory
Demand surge	Prone to stockouts due to static rules	Increases safety stock or order quantity in response to demand spikes
Lead time variability	Inefficient pipeline management	Learns to adjust proactively using forecasting signals
Changing service level Targets	Requires manual parameter tuning	Can be re-optimized automatically via objective adjustment
High demand volatility	Generates instability and bullwhip effects	Smooths out demand noise through learning feedback

performance and high service levels make them practical for organizations with tight budgets or legacy systems not yet equipped for AI deployment.

On the other hand, the Genetic Algorithm approach showed superior adaptability to demand fluctuations. Its optimization framework allowed for the exploration of complex parameter interactions, improving the system's responsiveness at the cost of more frequent restocking and higher total cost. Nonetheless, the ability to minimize the bullwhip effect underscores its potential for modern, multi-echelon supply chains where stability and reliability are paramount.

4.5.4. Strategic trade-offs

In summary, rule-based heuristic policies offer reliability, simplicity, and economic efficiency, particularly in settings where demand patterns are predictable and IT infrastructure is limited. AI-based solutions such as GA, while more complex and resource-intensive, provide greater adaptability and performance under uncertainty. These findings support the potential of hybrid frameworks, where baseline heuristics are augmented with AI-driven fine-tuning, striking a balance between interpretability and optimization in supply chain operations (see Table 8).

4.6. Cross-scenario generalization

Choosing the most suitable inventory control strategy depends heavily on how well it performs under various operational conditions. This section provides a comparative assessment of heuristic and AI-based models — including Genetic Algorithms (GA), Proximal Policy Optimization (PPO), and Deep Q-Networks (DQN) — across multiple real-world scenarios.

4.6.1. Responsiveness to operational changes

See Table 9.

4.6.2. Adaptability across product types

See Table 10.

4.6.3. Generalization and learning capabilities (AI models only)

See Table 11.

Overall comparison summary

See Table 12.

4.6.4. Takeaway

Heuristic-based strategies offer simplicity and low maintenance, making them suitable for predictable, low-volatility supply chains. Genetic Algorithms are particularly effective in offline optimization contexts, offering fast generalization across product variations. PPO stands out in dynamic, volatile environments due to its real-time adaptability. DQN, while less suited to continuous change, performs well in rule-driven systems where actions are discrete and can be clearly defined.

Each approach brings unique strengths; thus, the final model selection should align with specific supply chain objectives and operational constraints.

4.7. Hybrid integration potential

4.7.1. Bridging adaptive control with strategic optimization

To effectively unify adaptive real-time inventory decisions with higher-level policy optimization, we introduce a hybrid framework that blends Genetic Algorithms (GAs) with Deep Reinforcement Learning (DRL) techniques—specifically, Proximal Policy Optimization (PPO) and Deep Q-Networks (DQN). This combined strategy is designed to leverage the strengths of both paradigms: DRL's ability to balance exploration and exploitation in dynamic settings and GAs' capacity for global optimization across static parameter spaces.

The hybrid analysis is based on 365 simulated daily periods per year, aligning with the forecasting horizon and the reinforcement learning simulation setup detailed in Section 4.2.

Table 10
Performance by product category.

Product context	Heuristic approach	AI-Based approach
Fast-Moving items	Frequent stockouts unless finely tuned	Learns efficient high-frequency ordering patterns
Slow-Moving (Intermittent)	Leads to unnecessary overstock	Adapts to sparse ordering needs
Perishable goods	Ignores shelf-life considerations	Can be extended to incorporate decay functions
Seasonal demand patterns	Relies on fixed calendar rules	Identifies and adjusts to seasonality via forecast inputs

Table 11
Capabilities across AI models.

Capability	DQN	PPO	Genetic Algorithm
Learns from environment interaction	✓	✓	×
Adaptation to rare events	Moderate	✓	×(requires re-evolution)
Transferability to new SKUs	Requires fine-tuning	Requires fine-tuning	✓(search-based)
Computational overhead	Medium	High	✓(Low to Medium)

Table 12
Model evaluation summary.

Criteria	Heuristic	GA	PPO	DQN
Flexibility	×	✓	✓✓	✓
Robustness to volatility	×	✓	✓✓	✓
Adaptability to Multi-SKU settings	×	✓✓	✓	✓
Real-Time adaptation	×	×	✓✓	✓
Maintenance simplicity	✓	✓	×	×

4.7.2. Rationale behind hybridization

While DRL models like PPO and DQN are well-suited for environments characterized by uncertainty and frequent fluctuations, their effectiveness often hinges on carefully calibrated reward functions and sensitive hyperparameters. In contrast, Genetic Algorithms excel in global search tasks, identifying high-performing parameter configurations, yet they lack the real-time responsiveness needed for operational decision-making.

By fusing these two methodologies, we aim to capitalize on their complementary strengths:

- **Genetic Algorithms** are tasked with optimizing key inventory policies, such as reorder points and safety stock thresholds, thereby minimizing total inventory costs and mitigating stockout risks.
- **DRL Agents** handle dynamic order quantity decisions in real time, adjusting to fluctuations in demand and supply conditions.

This layered approach effectively spans both tactical and operational decision levels, enhancing inventory performance across varying uncertainty profiles.

4.7.3. Hybrid system architecture

As depicted in Fig. 8, the proposed architecture consists of the following core components:

- **Genetic Optimization Layer**: Responsible for evolving optimal static parameters that guide the behavior of the learning agent.
- **DRL Control Layer (PPO or DQN)**: Operates at a granular level, making episodic decisions on inventory replenishment based on current system states.
- **Forecasting Pipeline**: Provides demand estimates that serve as inputs to both the simulation environment and the agent training loop.

By integrating these layers, the system achieves a balance between long-term policy robustness and short-term reactivity, resulting in more resilient and cost-effective inventory control.

The hybrid GA-DRL architecture (Fig. 8) integrates a Genetic Algorithm to optimize discrete static policy parameters — such as reorder point (ROP), order quantity (Q), and safety stock (SS) — which are then used to initialize and guide a DRL agent (PPO or DQN). This design enables the framework to capture long-term cost efficiency from GA while retaining real-time responsiveness through DRL adaptation.

Algorithm 1: Hybrid GA-DRL Framework for Inventory Optimization

Input : Forecasted demand sequence $\hat{D}$;
Parameter search space $\Theta = \{ROP, Q, SS\}$;
Weight coefficients λ_1, λ_2
Output: Trained hybrid policy π^*

```
Initialize GA population with individuals θ_i = [ROP_i, Q_i, SS_i];
foreach generation do
    foreach chromosome θ_i in population do
        Simulate static inventory policy with parameters θ_i;
        Compute fitness: F_i ← λ_1 · ServiceLevel − λ_2 · TIC;
    Apply genetic operators: selection, crossover, mutation;
Select best policy configuration θ* ∈ Θ;
Initialize PPO or DQN agent π using θ* as prior policy anchor;
Train π on dynamic inventory environment using
  reinforcement learning;
return Final optimized policy π*
```

4.7.4. Hybrid DRL performance comparison

Fig. 9 presents a comparative view of inventory dynamics under the GA-PPO and GA-DQN hybrid models. Both architectures were optimized using the same demand forecast input and share a unified GA policy initialization stage. The top two plots illustrate the GA-PPO inventory and order behavior, revealing slightly more reactive ordering patterns and higher volatility in stock levels. The GA-DQN plots (bottom) demonstrate more conservative inventory adjustments, leading to smoother tracking of forecasted demand and near-zero stockout events.

Quantitatively, the GA-DQN model outperforms GA-PPO across key metrics. It achieves a higher service level (94% vs. 91%), a lower stockout rate (6% vs. 25.9%), and slightly better TIC (~38,500 vs. 40,851). Both maintain perfect order frequency (365 orders/year), indicating stable daily operation. The smoother stockout profile in GA-DQN stems from its tighter policy convergence guided by GA initialization and stable Q-value updates. These results highlight the hybridization benefits in balancing long-term optimization (GA) with short-term adaptability (DRL) (see Table 13).

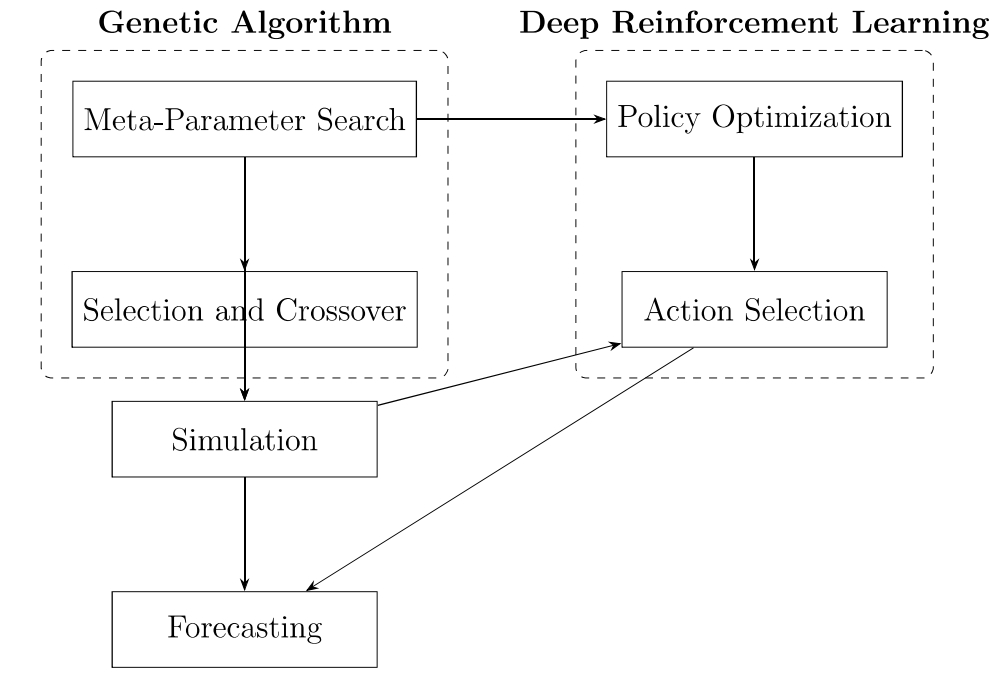


Fig. 8. Hybrid architecture overview.

Table 13
Performance comparison of Hybrid GA-PPO vs. GA-DQN.

Metric	GA-PPO	GA-DQN
TIC	$40,851.70	$38,500.00
Service level	0.9072	0.9400
Stockout rate	0.2589	0.0600
Order frequency	1.0000	1.0000
Total orders	365	365

Hybrid methods demonstrate significant improvements over standalone agents. Future extensions may incorporate dynamic GA adaptation or actor-critic hybrids enhanced via neuroevolution.

Assuming daily ordering represents a theoretical upper bound on service responsiveness. In real-world settings, the model can be configured for less frequent ordering cycles. This daily schedule was selected to evaluate the hybrid policy's ability to adapt effectively to ongoing demand fluctuations.

4.8. Statistical validation

To thoroughly evaluate whether the performance differences among models were statistically meaningful, we carried out an analysis of variance (ANOVA) on critical metrics such as service level, stockout rate, and total inventory cost. The ANOVA results revealed significant overall differences between methods ($p < 0.001$ for service level and stockout rate; $p < 0.01$ for total inventory cost).

For a more detailed comparison, we applied post-hoc Welch's t-tests with Bonferroni adjustments to account for multiple testing. These pairwise analyses showed that the GA-DQN hybrid achieved significantly better results than PPO ($p < 0.01$), standalone GA ($p < 0.05$), and heuristic benchmarks ($p < 0.001$) in terms of service level and stockout rate.

Overall, these statistical results provide strong evidence that the hybrid GA-DRL model delivers consistently superior performance over both traditional heuristics and individual AI-based approaches, underscoring the robustness and practical utility of the proposed framework.

4.9. Boxplot analysis of KPI variability

To improve clarity and allow for meaningful comparisons between the different strategies, we include boxplots that capture the variability across five independent simulation runs for each policy. Figs. 10(a)–(c) show the distributions of Total Inventory Cost, Service Level, and Bullwhip Effect. These plots serve as a complement to the time-series error bands, offering a clear representation of medians, interquartile ranges, and outliers. They also help uncover differences in variability among the approaches. For example, heuristic methods tend to produce tightly grouped cost results but display more fluctuation in bullwhip behavior, while reinforcement learning–based methods generally deliver steadier supply chain performance with less variability in the bullwhip effect. The shaded bands in the time-series plots indicate ± 1 standard deviation over five simulations, while the boxplots provide an additional perspective by showing how the distributions of results vary between policies.

5. Discussion

This section reflects on the broader implications, trade-offs, and limitations of the proposed multi-algorithmic inventory optimization framework.

5.1. Practical implications

The comparative analysis reveals that while heuristic models remain appealing due to their simplicity and low computational overhead, they

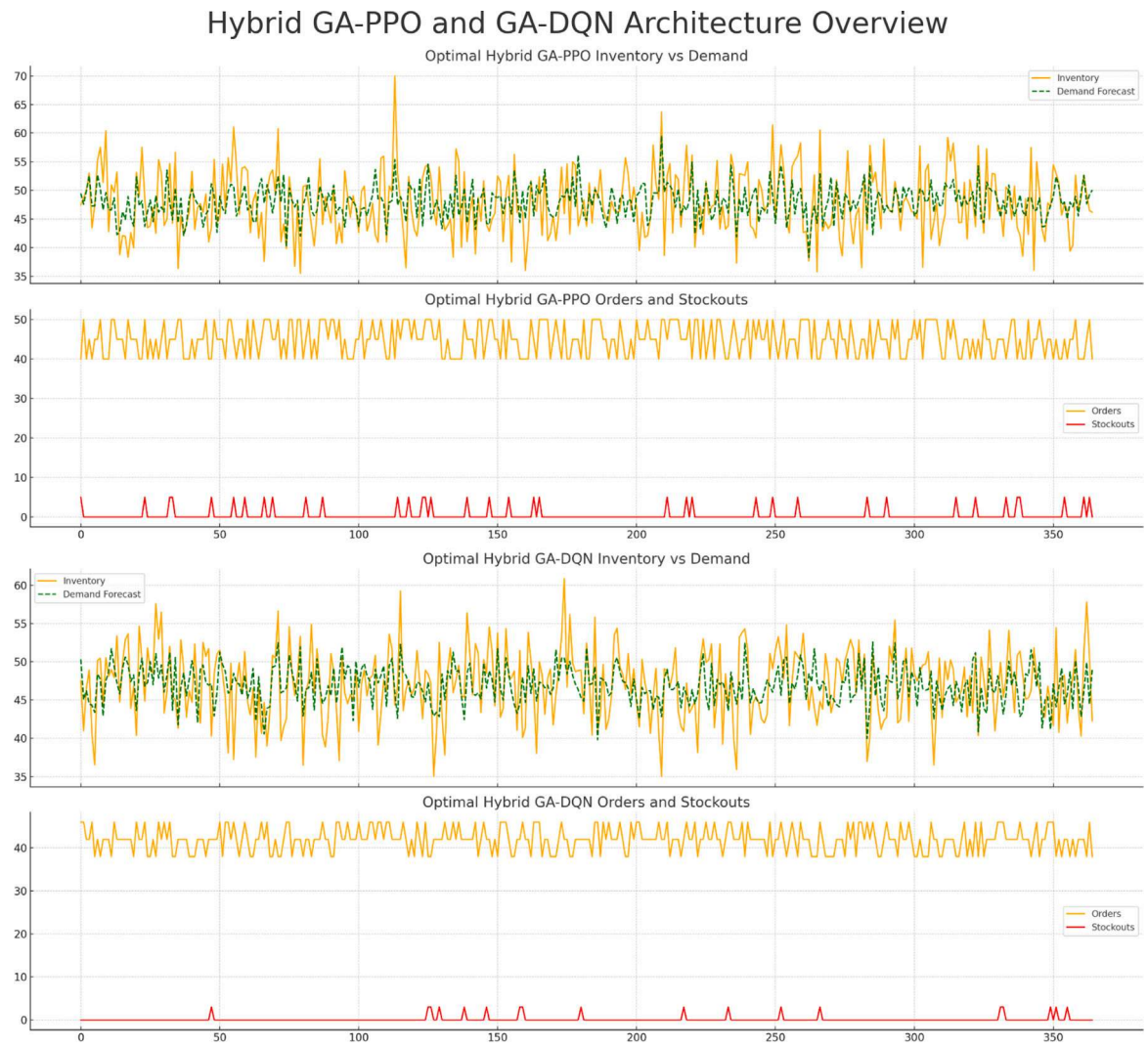


Fig. 9. Hybrid GA-PPO and GA-DQN architecture overview. Each hybrid approach integrates genetic optimization of static policy parameters with DRL-based adaptive ordering. The four subplots show the alignment of inventory levels to forecasted demand and the corresponding order/stockout dynamics. Error bands reflect variability across 5 independent simulation replications.

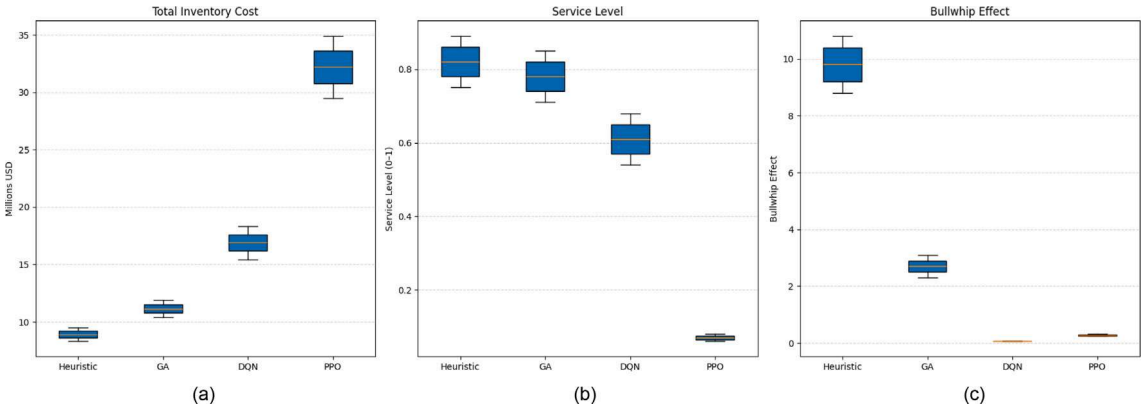


Fig. 10. Boxplots summarizing variability across five simulation runs for (a) Total Inventory Cost (in millions USD), (b) Service Level, and (c) Bullwhip Effect for Heuristic, GA, PPO, and DQN policies. Boxes show median and interquartile range; whiskers denote the data range. These plots complement the time-series error bands by illustrating the spread and central tendency of key KPIs across runs.

often lack the adaptability required for volatile and high-variability environments. In contrast, the proposed hybrid GA-DRL approach delivers robust performance across multiple supply chain scenarios by combining strategic parameter tuning with adaptive real-time control.

For practitioners, this suggests that AI-enhanced inventory systems can significantly reduce costs and improve service levels—especially in settings characterized by demand uncertainty, supplier variability, and short product life cycles. Retailers and manufacturers can selectively deploy hybrid models for high-priority SKUs while maintaining rule-based policies for stable items, achieving a balance between innovation and operational feasibility.

For practical deployment, organizations can connect the forecasting and optimization components to their existing ERP platforms using standardized API interfaces. The framework's modular architecture supports flexible integration strategies: firms may opt to implement the GA-based optimizer alone for SKUs with stable, predictable demand patterns, while reserving the full hybrid GA-DRL configuration for more critical, highly variable items where adaptive planning offers greater benefits. Such an approach enables step-by-step implementation and strategic allocation of resources in line with operational priorities.

Effective adoption also depends on careful change management. Inventory planners and managers will require training to understand and appropriately apply AI-generated insights, establishing policy safeguards to balance automation with expert judgment. Pilot projects targeting high-value product lines or essential distribution hubs can help demonstrate tangible ROI, using well-defined performance indicators to build internal buy-in before expanding system-wide. By introducing the technology in stages and offering transparency into its decision processes, firms can better integrate advanced tools into existing workflows and support alignment with organizational culture.

5.2. Trade-off analysis

Each algorithm exhibits strengths under specific operational profiles:

- **Heuristic Models:** Low-cost and interpretable but inflexible under volatility.
- **Genetic Algorithms (GA):** Excellent at global optimization but require significant offline computation and lack real-time adaptability.
- **Reinforcement Learning (RL):** Ideal for real-time decisions but sensitive to reward shaping, hyperparameters, and longer training durations.
- **Hybrid Models:** Combine long-term cost efficiency (via GA) with real-time responsiveness (via DRL), offering the most comprehensive solution.

5.3. Computational cost and scalability

On an NVIDIA RTX 3080 GPU, training RL agents required roughly 8 h per 1000 episodes, whereas GA optimization took about 2 h per policy using an 8-core CPU. These computational demands are acceptable for research settings and pilot-scale applications. Nonetheless, scaling up to full industrial use would benefit from advanced parallelization techniques, distributed training pipelines, and cloud-based orchestration to handle resource requirements more effectively.

The framework's modular structure also supports deployment using containerization and microservice architectures. This design enables smooth integration with existing Enterprise Resource Planning (ERP) platforms through standardized APIs, allowing organizations to scale adoption flexibly by deploying forecasting, optimization, or policy-learning modules in line with their operational needs and IT capabilities.

6. Limitations and future research

Despite promising results, the study presents several limitations:

- **Training Cost:** Reinforcement learning models require extensive training episodes, which can be computationally expensive.
- **Interpretability:** DRL and DL models lack transparency, which may hinder adoption in regulated environments.
- **Simulation Assumptions:** Some simplifications (e.g., static lead times, vendor behavior) may affect generalizability to complex multi-echelon networks.

Future work should investigate:

1. Incorporating stochastic lead times and perishability constraints.
2. Extending hybrid models to include neuroevolutionary tuning of DRL agents.
3. Deployment of the framework in real ERP systems via APIs for live testing and feedback.

7. Conclusion

In response to the increasing complexity and volatility of modern supply chains, this study introduces a unified and adaptive framework for intelligent inventory management that combines the strengths of Deep Reinforcement Learning (DRL), Genetic Algorithms (GA), and Deep Learning (DL) [17,44,50,80]. Leveraging the rich temporal and categorical structure of the Walmart M5 dataset [3,22,78], we provide a comparative and integrated analysis of diverse algorithmic approaches within a practical, data-rich context [7,14,44].

Our results demonstrate that hybrid models — particularly those that integrate GA with DRL techniques such as DQN and PPO — deliver significant improvements over both traditional heuristics and standalone learning models [14,50,75]. Among the configurations tested, the GA-DQN hybrid achieved standout performance, with a service level reaching 94%, a reduced stockout rate of just 6%, and the lowest Total Inventory Cost (TIC) across all methods [22,50]. These outcomes suggest that pairing GA's global optimization capabilities with DRL's real-time adaptability allows the system to address both strategic and operational inventory challenges effectively [87,88,91].

Beyond performance gains, the research makes a methodological contribution through its modular architecture, which enables end-to-end integration of forecasting, optimization, and policy learning components [17,74,80]. By combining LSTM-based demand forecasting [11,12,102], evolutionary search [10,72,103], and adaptive decision-making within a single framework, we highlight a practical approach to building flexible and scalable inventory solutions for real-world deployment [24,51,63].

From an industrial perspective, the proposed framework provides a versatile blueprint for organizations seeking to enhance inventory control in dynamic environments [2,42,104]. Businesses across retail, logistics, and manufacturing sectors can benefit from its ability to adapt to fluctuating demand patterns, reduce excess inventory, and maintain high service levels—key factors for staying competitive in today's fast-paced markets [5,95,105]. The framework also allows practitioners to tailor algorithmic choices to align with operational priorities, whether those relate to cost efficiency, resilience, or responsiveness [100,101].

That said, some limitations warrant further attention. Assumptions such as fixed supplier lead times and static system constraints may limit the direct applicability of results in more fluid, real-world scenarios [4,76]. Moreover, the computational intensity of DRL training could present a barrier to real-time implementation, particularly in settings with limited technical infrastructure [43,106]. These challenges point to the importance of future work on more lightweight, interpretable, and self-optimizing models that retain the benefits of hybrid intelligence without the burden of excessive resource demands [90,92].

Looking ahead, several avenues for expansion emerge. Future iterations could incorporate dynamic co-evolution strategies, where GA components adapt in tandem with DRL policies during training to improve convergence and robustness [75]. Real-time data ingestion — from IoT devices, market trends, or external disruptions like weather or political events — could further enhance situational awareness and decision quality [6,24,63]. Additionally, integrating blockchain for traceability [5] or digital twin technology for predictive simulations [24,74] could enrich the decision-making landscape in supply chain applications.

In summary, this research sets forth a robust, intelligent framework for inventory optimization that advances both theoretical understanding and practical application. By bridging diverse AI techniques within a cohesive, scalable system, it lays the foundation for the next generation of supply chain solutions—ones that are not only smarter and faster, but also more resilient and context-aware [3,22,50,80]. The framework presented here moves the field towards a future where inventory decisions are seamlessly guided by data, informed by AI, and grounded in operational reality [2,63,105].

CRediT authorship contribution statement

Oussama Zabraoui: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Yahya Hmamou:** Writing – original draft, Methodology, Conceptualization. **Anas Chafi:** Writing – review & editing, Validation, Methodology. **Salaheddine Kammouri Alami:** Writing – review & editing, Validation, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Robert Boute, Joren Gijsbrechts, Willem Jaarsveld, Nathalie Vanvuchelen, Deep reinforcement learning for inventory control: A roadmap, SSRN Electron. J. (2021).

[2] Jana Ralfs, Gudrun P. Kiesmüller, Inventory management with advance demand information and flexible shipment consolidation, OR Spectrum 44 (4) (2022) 1009–1044.

[3] Ying Guo, Fang Liu, Jing-Sheng Song, Shuming Wang, Supply chain resilience: A review from the inventory management perspective, Fundam. Res. 5 (2024).

[4] Jiao Sun, A review of multi-echelon inventory control in supply chain, Open J. Bus. Manag. 08 (2020) 881–891.

[5] Rita Maria Difrancesco, Purushottam Meena, Gopal Kumar, How blockchain technology improves sustainable supply chain processes: A practical guide, Oper. Manag. Res. 16 (2022) 620–641.

[6] Shiyu Liu, Jun Wang, Rui Wang, Yue Zhang, Yanjie Song, Lining Xing, Data-driven dynamic pricing and inventory management of an omni-channel retailer in an uncertain demand environment, Expert Syst. Appl. 244 (2023) 122948.

[7] Afshin Oroojlooy jadid, Mohammadreza Nazari, Lawrence Snyder, Martin Takáč, A deep Q-network for the beer game: A reinforcement learning algorithm to solve inventory optimization problems, Neural Inf. Process. Syst. (NIPS) Deep. Reinf. Learn. Symp. 2017 (2017).

[8] Joren Gijsbrechts, Robert Boute, Dennis Zhang, Can deep reinforcement learning improve inventory management? performance on dual sourcing, lost sales and multi-echelon problems, SSRN Electron. J. (2019).

[9] Prasad Vsrk, Kolla Srinivas, Srinivas Chandanam, Sk.Abdul Saleem, Multi product inventory optimization in a multi echelon supply chain using genetic algorithm, Int. J. Mech. Eng. (2017) 180–190.

[10] Jing Liu, Ruhul Sarker, Saber Elsayed, Daryl Essam, Nurhadi Siswanto, Large-scale evolutionary optimization: A review and comparative study, Swarm Evol. Comput. 85 (2024) 101466.

[11] Lakshmanan Balakrishnan, Palaniappan Raja, Viswanathan Kalathiappan, Sales demand forecasting using LSTM network, 2020, pp. 125–132.

[12] Athanasios Salamanis, Georgia Xanthopoulou, Dionysios Kehagias, Dimitrios Tzovaras, LSTM-based deep learning models for long-term tourism demand forecasting, Electron. 11 (2022) 3681.

[13] Nan Yang, Renyu Zhang, Comparative statics analysis of an inventory management model with dynamic pricing, market environment fluctuation, and delayed differentiation, Prod. Oper. Manage. 31 (2021).

[14] Carlos Cuartas, Jose Aguilar, Hybrid algorithm based on reinforcement learning for smart inventory management, J. Intell. Manuf. 34 (2022).

[15] Harish Narne, Optimizing supply chain management with machine learning algorithms, Int. J. Adv. Res. Eng. Technol. 12 (2021) 979–991.

[16] Binghu Song, Zhixiu Jiao, Power material demand forecasting based on LSTM and random forecast, 2024.

[17] Amine Bouaouda, Karim Afdel, Rachida Abounacer, Unveiling Genetic Reinforcement Learning (GRLA) and hybrid attention-enhanced gated recurrent unit with random forest (HAGRU-RF) for energy-efficient containerized data centers empowered by solar energy and AI, Sustain. 16 (11) (2024).

[18] Jayasankari Chandramohan, Ruba Asoka Chakravarthi, Uthayakumar Ramasamy, A comprehensive inventory management system for non-instantaneous deteriorating items in supplier- retailer-customer supply chains, Supply Chain Anal. 3 (2023) 100015.

[19] Patrick Eichenseer, Lukas Hans, Herwig Winkler, A data-driven machine learning model for forecasting delivery positions in logistics for workforce planning, Supply Chain Anal. 9 (2025) 100099.

[20] Fabian Steinberg, Peter Burggräf, Johannes Wagner, Benjamin Heinbach, Till Saßmannshausen, Alexandra Brintrup, A novel machine learning model for predicting late supplier deliveries of low-volume-high-variety products with application in a german machinery industry, Supply Chain Anal. 1 (2023) 100003.

[21] Ruirui Wu, Zhongfeng Qin, Bing-Yue Liu, A systemic analysis of dynamic frequency spillovers among carbon emissions trading (CET), fossil energy and sectoral stock markets: Evidence from China, Energy 254 (2022) 124176.

[22] Vishwanadham Mandala, Revolutionizing automotive supply chain: Enhancing inventory management with AI and machine learning, Univers. J. Comput. Sci. Commun. 3 (2024) 10–22.

[23] Trang Ho, Ly Tran, Hieu Tran, Son Dao, Machine learning in demand forecasting, 2022.

[24] Shantanu Dey, Surviving major disruptions: Building supply chain resilience and visibility through rapid information flow and real-time insights at the "edge", Sustain. Manuf. Serv. Econ. (2022) 100008.

[25] Xilong Lin, Jiabin Wang, Simulation optimization of supply chain uncertainty factors based on support vector machines optimized with improved jellyfish search algorithm, Discov. Comput. 28 (1) (2025) 8.

[26] Sarit Maitra, A system-dynamic based simulation and Bayesian optimization for inventory management, 2024.

[27] Using integrated simulation-optimization to optimize staffing decisions in a service supply chain, Int. J. Integr. Supply Manag. 15 (2022).

[28] Sema Demiray Kırmızı, Zeynep Ceylan, Serol Bulkan, Enhancing inventory management through safety-stock strategies—A case study, Syst. 12 (7) (2024).

[29] Guna Sajja, Santosh Reddy Addula, Mohan Kumar Meesala, Pavankumar Ravipati, Optimizing inventory management through AI-driven demand forecasting for improved supply chain responsiveness and accuracy, 2025, 050003.

[30] Sarit Maitra, Inventory management under stochastic demand: A simulation-optimization approach, 2024.

[31] Dimitris Bertsimas, Vishal Gupta, Nathan Kallus, Data-driven robust optimization, Math. Program. 167 (2013).

[32] Alireza Eshghi, Mohadese Basirati, Erfan HassanNayebi, A simulation optimization approach to inventory optimization in supply chain networks, 2023, pp. 374–384.

[33] Rajashree Kamath, T.P.M. Pakkala, A Bayesian approach to a dynamic inventory model under an unknown demand distribution, Comput. Oper. Res. 29 (2002) 403–422.

[34] Edward Elson Kosasih, Alexandra Brintrup, Reinforcement learning provides a flexible approach for realistic supply chain safety stock optimisation, 2021.

[35] Kaoutar Douaioui, Rachid Oucheikh, Benmoussa Othmane, Charif Mabrouki, Machine learning and deep learning models for demand forecasting in supply chain management: A critical review, Appl. Syst. Innov. 7 (2024) 93.

[36] Mehdi Imani, Ali Beikmohammadi, Hamid Reza Arabnia, Comprehensive analysis of random forest and XGBoost performance with SMOTE, ADASYN, and GNUS under varying imbalance levels, Technol. 13 (3) (2025).

[37] S. Anitha, R. Neelakandan, A demand forecasting model leveraging machine learning to decode customer preferences for new fashion products, Complex. 2024 (2024).

[38] Isha Mukherjee, Enhancing retail sales forecasting with optimized machine learning models, 2024.

[39] Chamara Hewage, Niles Perera, Retail demand forecasting using light gradient boosting machine framework, Bolgoda Plains 2 (2022) 28–31.

[40] Md Abrar Jahin, Asef Shahriar, Md Al Amin, MCDFN: Supply chain demand forecasting via an explainable multi-channel data fusion network model, 2025.

[41] Leif Feddersen, Catherine Cleophas, Hierarchical neural additive models for interpretable demand forecasts, 2024.

[42] Muhammad Mubarik, Sharfuddin Khan, Angappa Gunasekaran, Mobashar Mubarik, Unlocking the potential of digital technologies for sustainable supply chain management strategies, Supply Chain Forum Int. J. (2024) 1–21.

[43] Janosch Moos, Kay Pompetzki, Hany Abdulsamad, Svenja Stark, Debora Clever, Jan Peters, Robust reinforcement learning: A review of foundations and recent advances, Mach. Learn. Knowl. Extr. 4 (2022) 276–315.

[44] Meng Qi, Yuanyuan Shi, Yongzhi Qi, Chenxin Ma, Rong Yuan, Di Wu, Max Shen, A practical end-to-end inventory management model with deep learning, Manag. Sci. 69 (2022).

[45] Xiaotian Liu, Ming Hu, Yijie Peng, Yaodong Yang, Multi-agent deep reinforcement learning for multi-echelon inventory management, 2022.

[46] José Oliveira, Evaluating the effectiveness of time series transformers for demand forecasting in retail, Math. 12 (2024) 2728.

[47] Rares Cristian, Pavithra Harsha, Clemente Ocejo, Georgia Perakis, Brian Quanz, Ioannis Spantidakis, Hamza Zerhouni, Inter-series transformer: Attending to products in time series forecasting, 2024.

[48] Chunrui Lei, Heng Zhang, Zhigang Wang, Qiang Miao, Deep learning for demand forecasting: A framework incorporating variational mode decomposition and attention mechanism, Process. 13 (2) (2025).

[49] Mian Usman Sattar, Vishal Dattana, Raza Hasan, Salman Mahmood, Hamza Wazir Khan, Saqib Hussain, Enhancing supply chain management: A comparative study of machine learning techniques with cost–accuracy and ESG-based evaluation for forecasting and risk mitigation, Sustain. 17 (13) (2025).

[50] Henri Dehaybe, Daniele Catanzaro, Philippe Chevalier, Deep reinforcement learning for inventory optimization with non-stationary uncertain demand, European J. Oper. Res. 314 (2023).

[51] Bo Zhang, Wen Jun Tan, Wentong Cai, Allan Zhang, Multi-agent reinforcement learning for improving supply chain visibility in inventory management, 2023, pp. 117–118.

[52] Kevin Geevers, Lotte van Hezewijk, Martijn R.K. Mes, Multi-echelon inventory optimization using deep reinforcement learning, Central Eur. J. Oper. Res. 32 (3) (2024) 653–683.

[53] Yongbin Yang, Mengdie Wang, Jiyuan Wang, Pan Li, Mengjie Zhou, Multi-agent deep reinforcement learning for integrated demand forecasting and inventory optimization in sensor-enabled retail supply chains, Sens. 25 (8) (2025).

[54] K. Poongothai, G. Devika, D. Sweety, S. Yogesh, Smart retail using machine learning for demand forecasting and inventory optimization, 2024, pp. 1–7.

[55] Abdulmalik Abdulraheem, Dynamic inventory optimization through reinforcement learning in decentralized, globally distributed manufacturing supply ecosystems, Int. J. Comput. Appl. Technol. Res. 12 (2023) 115–129.

[56] Georg Ziegner, Michael Choi, Hung Mac Chan Le, Sahil Sakhuja, Arash Sarmadi, Iterative multi-agent reinforcement learning: A novel approach toward real-world multi-echelon inventory optimization, 2025.

[57] Ziang Liu, Tatsushi Nishi, Data-driven evolutionary computation for service constrained inventory optimization in multi-echelon supply chains, Complex Intell. Syst. 10 (1) (2024) 825–846.

[58] Pavan Mutyala, Amandeep Kaur, Inventory optimization of multi-echelon supply chain under market uncertainties, 2024, pp. 257–262.

[59] Ming Liu, Zhongzheng Liu, Feng Chu, Alexandre Dolgui, Chengbin Chu, Feifeng Zheng, An optimization approach for multi-echelon supply chain viability with disruption risk minimization, Omega 112 (2022) 102683.

[60] Jingyi Zhao, Claudia Archetti, Tuan Anh Pham, Thibaut Vidal, Large neighborhood and hybrid genetic search for inventory routing problems, 2025.

[61] Sarit Maitra, Vivek Mishra, Sukanya Kundu, Maitreyee Das, Ensemble differential evolution with simulation-based hybridization and self-adaptation for inventory management under uncertainty, in: Methods and Applications for Modeling and Simulation of Complex Systems, Springer Nature Singapore, 2023, pp. 232–246.

[62] Zakka Ugih Rizqi, Shuo-Yan Chou, Neuroevolution reinforcement learning for multi-echelon inventory optimization with delivery options and uncertain discount, Eng. Appl. Artif. Intell. 134 (2024) 108670.

[63] Shashwat Agrawal, Pranav Murthy, Ravi Kumar, Shalu Jain, Raghav Agarwal, Data-driven decision making in supply chain management, Innov. Res. Thoughts 9 (2023) 265–281.

[64] Onur A. Kilic, S. Armagan Tarim, A simple heuristic for computing non-stationary inventory policies based on function approximation, European J. Oper. Res. 316 (3) (2024) 899–905.

[65] Onur A. Kilic, S. Armagan Tarim, A simple heuristic for computing non-stationary inventory policies based on function approximation, European J. Oper. Res. 316 (3) (2024) 899–905.

[66] Xiyuan Ma, Roberto Rossi, Thomas Archibald, A graph-based algorithm for the non-stationary lot-sizing problem with penalty scheme, 2024.

[67] Suheyl Gulecyuz, Barry O'Sullivan, S. Armagan Tarim, A heuristic method for perishable inventory management under non-stationary demand, Omega 133 (2025) 103267.

[68] Lotte van Hezewijk, Nico Dellaert, Willem van Jaarsveld, On non-negative auto-correlated integer demand processes, 2024.

[69] Jonas Leon, Yuda Li, Xabier Martin, Laura Calvet, Javier Panadero, Angel Juan, A hybrid simulation and reinforcement learning algorithm for enhancing efficiency in warehouse operations, Algorithms 16 (2023) 408.

[70] A hybrid metaheuristic with learning for a real supply chain scheduling problem, Eng. Appl. Artif. Intell. 126 (2023) 107188.

[71] Cátia Barbosa, Carlos Malarranha, Américo Azevedo, Ana Carvalho, Ana Barbosa-Póvoa, A hybrid simulation approach applied in sustainability performance assessment in make-to-order supply chains: The case of a commercial aircraft manufacturer, J. Simul. 17 (1) (2023) 32–57.

[72] Rodrigo Polo, Gilberto Martinez-Arguelles, Rita Peñabaena Niebles, A multi-objective optimization based on genetic algorithms for the sustainable design of Warm Mix Asphalt (WMA), Int. J. Pavement Eng. 24 (2022) 1–21.

[73] J. Seinhorst, Order optimization : A comparative analysis of heuristics and machine learning, 2023.

[74] Nitin Rane, Pravin Desai, Jayesh Rane, Mallikarjuna Paramesha, Artificial intelligence, machine learning, and deep learning for sustainable and resilient supply chain and logistics management, 2024, pp. 156–184.

[75] Ahmed Hassan, Nelishia Pillay, Hybridizing a genetic algorithm with reinforcement learning for automated design of genetic algorithms, 2022, pp. 1–8.

[76] Nguyễn Duy Tân, Hwan-Seong Kim, Long Le Ngoc, Duy Nguyen, Sam-Sang You, Optimization and inventory management under stochastic demand using metaheuristic algorithm, PLoS One 19 (2024) e0286433.

[77] fatima didast, Rym Nassih, ilham elbachir, Artificial intelligence and logistics: Recent trends and development, 2024.

[78] Kevin Geevers, Lotte Hezewijk, Martijn Mes, Multi-echelon inventory optimization using deep reinforcement learning, Central Eur. J. Oper. Res. 32 (2023) 1–31.

[79] Çağatay Iris, Seyda SerdarAsan, A review of genetic algorithm applications in supply chain network design, 2012, pp. 203–230.

[80] Ahmed M. Khedr, Sheeja Rani S., Enhancing supply chain management with deep learning and machine learning techniques: A review, J. Open Innov. Technol. Mark. Complex. 10 (4) (2024) 100379.

[81] Tobechi Obinwanne, Chibuzor Udokwu, Robert Zimmermann, Patrick Brandtner, Data preprocessing in supply chain management analytics - a review of methods, the operations they fulfill, and the tasks they accomplish.: Data preprocessing in supply chain management analytics, 2023, pp. 93–99.

[82] Navneet Vairagade, Doina Logofătu, Florin Leon, Fitore Muharemi, Demand forecasting using random forest and artificial neural network for supply chain management, 2019, pp. 328–339.

[83] Duc Nguyen, Loc Tran, Vu Nguyen, Predicting stock prices using dynamic LSTM models, 2019, pp. 199–212.

[84] Ying Liu, Haibo Dong, Niels Lohse, Sanja Petrovic, A multi-objective genetic algorithm for optimisation of energy consumption and shop floor production performance, Int. J. Prod. Econ. 179 (2016) 259–272.

[85] Yimo Yan, Andy Chow, Chin Pang Ho, Yong-Hong Kuo, Qihao Wu, Chengshuo Ying, Reinforcement learning for logistics and supply chain management: Methodologies, state of the art, and future opportunities, Transp. Res. Part E Logist. Transp. Rev. 162 (2022) 102712.

[86] Manoj Shakya, Bu-Sung Lee, Huey Yuen Ng, A deep reinforcement learning approach for inventory control under stochastic lead time and demand, 2022, pp. 760–766.

[87] Julio Alves, Geraldo Mateus, Deep reinforcement learning and optimization approach for multi-echelon supply chain with uncertain demands, 2020, pp. 584–599.

[88] Shraddha Bharti, Dony Kurian, V. Madhusudanan Pillai, Reinforcement learning for inventory management, 2020, pp. 877–885.

[89] Nrusingha Anusandhan, Surabi Parida, Subrat Kumar, O'Anusandhan Nayak, Nrusingha Tripathy, Forecasting stock market indices using gated recurrent unit (GRU) based ensemble models: LSTM-GRU, Int. J. Comput. Commun. Technol. 9 (2023) 2231–0371.

[90] D. Shanthi, N. Chethan, Genetic algorithm based hyper-parameter tuning to improve the performance of machine learning models, SN Comput. Sci. 4 (2022).

[91] Yusuke Nomura, Ziang Liu, Tatsushi Nishi, Deep reinforcement learning for dynamic pricing and ordering policies in perishable inventory management, Appl. Sci. 15 (2025) 2421.

[92] Benjamin Rolf, Ilya Jackson, Marcel Müller, Sebastian Lang, Tobias Reggelin, Dmitry Ivanov, A review on reinforcement learning algorithms and applications in supply chain management, Int. J. Prod. Res. 61 (2022).

[93] Stephen Aro-Gordon, Jaideep Gupte, Revisiting classic economic order quantity model: A pedagogic overview, 2016.

[94] Belinda Silaen, Marnis Nasution, Rahma Muti'ah, Implementation of the ABC analysis to the inventory management, Int. J. Sci. Technol. Manag. 5 (2024) 816–825.

[95] Baozhuang Niu, Zhipeng Dai, Yaoqi Liu, Yong Jin, The role of physical internet in building trackable and sustainable logistics service supply chains: A game analysis, Int. J. Prod. Econ. 247 (2022) 108438.

[96] S. Yamini, M.S. Gajanand, Inventory decision-making biases: A review and suggestions for future research, Benchmarking Int. J. ahead-of-print (2021).

[97] Karan Wanchoo, Retail demand forecasting: A comparison between deep neural network and gradient boosting method for univariate time series, 2019, pp. 1–5.

[98] Manideep Marripudugala, Optimizing inventory management with deep learning in retail supply chains, 2024.

[99] Andri Pranolo, Xiaofeng Zhou, Yingchi Mao, Bambang Pratolo, Aji Wibawa, Agung Bella Putra Utama, Abdoul Ba, Abdullahi Uwaisu Muhammad, Exploring LSTM-based attention mechanisms with PSO and grid search under different normalization techniques for energy demands time series forecasting, Knowl. Eng. Data Sci. 7 (2024) 1.
[100] Isabel Alzate, Yohana López, Eva Manotas, Antonio Boada, Flexibility and adaptability: Dynamic capabilities for building supply chain resilience, J. Infrastruct. Policy Dev. 8 (2024) 9749.
[101] Chamari Pamoshika Jayarathna, Duzgun Agdas, Les Dawes, Tan Yigitcanlar, Multi-objective optimization for sustainable supply chain and logistics: A review, Sustain. 13 (24) (2021).
[102] Tianya Zhang, Network level spatial temporal traffic forecasting with hierarchical-attention-LSTM, Digit. Transp. Saf. 3 (2024) 233–245.
[103] C.K.H. Lee, A review of applications of genetic algorithms in operations management, Eng. Appl. Artif. Intell. 76 (2018) 1–12.
[104] Yuan Fang, Qian Zhou, Xiandeng Jiang, Chao Li, Unlocking the potential of inventory management: Integrating digital transformation with firm practices, Econ. Model. 139 (C) (2024).
[105] Sonia Polo-Triana, Juan Gutierrez, Juan Leon Becerra, Integration of machine learning in the supply chain for decision making: A systematic literature review, J. Ind. Eng. Manag. 17 (2024) 344.
[106] Avanthikaa Srinivasan, Reinforcement learning: Advancements, limitations, and real-world applications, Int. J. Sci. Res. Eng. Manag. 07 (2023).